CIVIL ENGINEERING PLANS FOR:
1319 MANGONIA PARK WAREHOUSE

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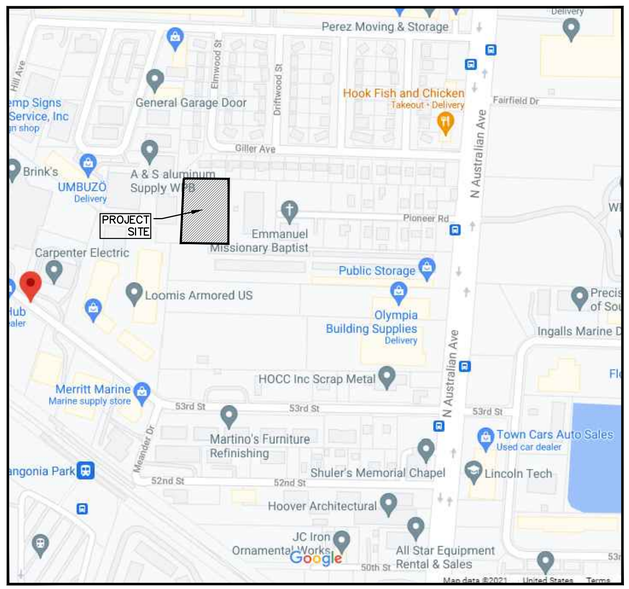
MANGONIA PARK, FLORIDA

MMA PROJECT #21-030

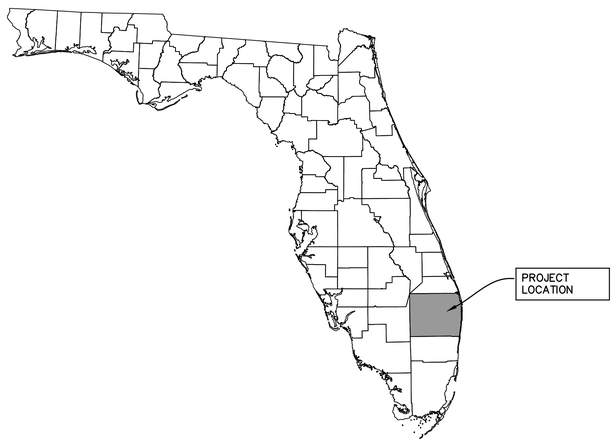
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GENERAL NOTES:

1. COORDINATION OF DRAWINGS AND DISCIPLINES. IT IS THE CONTRACTOR'S RESPONSIBILITY TO REVIEW ALL DRAWINGS AND SPECIFICATIONS FOR COMPLETENESS AND ACCURACY. FURTHER CONTRACTOR IS COORDINATE ALL TRADES AND DISCIPLINES PROPERLY. ANY DISCREPANCIES IN PLANS OR SPECIFICATIONS ARE TO BE BROUGHT TO THE ATTENTION OF THE ENGINEER IN WRITING.
2. TOWN R/W. ALL MATERIALS AND CONSTRUCTION WITHIN THE TOWN RIGHT-OF-WAY SHALL CONFORM TO THE LATEST FDOT DESIGN STANDARDS & LATEST SPECIFICATIONS FOR ROAD & BRIDGE CONSTRUCTION.
3. BUILDINGS & UTILITIES. CONTRACTOR IS ADVISED THAT THE BUILDINGS AND ASSOCIATED SYSTEMS ARE STILL UNDER REVIEW BY VARIOUS JURISDICTIONAL AGENCIES. AS SUCH, THERE MAY BE MODIFICATIONS TO SERVICE LINES (MATERIALS, SIZES, LOCATIONS, DEPTHS, ETC.) AS THE DOCUMENTS ARE PERMITTED.
4. SAFETY. IT SHALL BE THE CONTRACTOR'S SOLE RESPONSIBILITY TO TAKE THE NECESSARY PRECAUTIONS TO ENSURE PROPER SAFETY AND WORKMANSHIP WHEN WORKING IN THE VICINITY OF EXISTING UTILITY LINES. THE CONTRACTOR SHALL BE RESPONSIBLE TO COORDINATE WITH THE RESPECTIVE UTILITY PROVIDER ON ANY WORK IN THE VICINITY OF OVERHEAD OR UNDERGROUND LINES. CONTRACTOR SHALL VERIFY PROPER CLEARANCE BELOW EXISTING OVERHEAD LINES PRIOR TO WORKING WITHIN THE VICINITY OF THE LINES.
5. FAMILIARITY WITH SITE CONDITIONS. THE CONTRACTOR SHALL VISIT THE SITE AND ACQUAINT HIMSELF WITH ALL EXISTING CONDITIONS. MAKE HIS OWN SUBSURFACE INVESTIGATION TO SATISFY HIMSELF AS TO THE SITE AND SUBSURFACE CONDITIONS, BUT SUCH SUBSURFACE INVESTIGATIONS SHALL BE PERFORMED ONLY UNDER TIME SCHEDULES AND ARRANGEMENTS APPROVED IN ADVANCE BY THE OWNER'S REPRESENTATIVE AND ENGINEER. THE EXISTING GRADES SHOWN ON THE DRAWINGS ARE APPROXIMATE ONLY AND NO REPRESENTATION IS MADE AS TO THEIR ACCURACY OR CONSISTENCY. THE CONTRACTOR SHALL VERIFY ALL EXISTING GRADES TO THE EXTENT NECESSARY TO INSURE COMPLETION OF THE JOB TO THE PROPOSED GRADES INDICATED ON THE DRAWINGS.
6. WORKMANSHIP. ALL WORK SHALL BE PERFORMED IN A WORKMANSHIP-LIKE MANNER AND SHALL CONFORM WITH ALL APPLICABLE CITY, COUNTY, STATE AND FEDERAL REGULATIONS AND/OR CODES, INCLUDING OSHA.
7. ELEVATIONS. ALL ELEVATIONS REFER TO NORTH AMERICAN VERTICAL DATUM OF 1988 (NAVD 88).
8. SURVEY & BENCHMARK. REFER TO SURVEY PR[redacted]
9. GEOTECHNICAL INFORMATION. REFER TO GEOTECHNICAL REPORTS AND BORING LOGS FOR ADDITIONAL INFORMATION INCLUDING SITE PREPARATION REQUIREMENTS.
10. FIELD LOCATIONS. THE CONTRACTOR SHALL CONTACT "SUNSHINE STATE ONE CALL OF FLORIDA, INC. (1-800-432-4770)" AT LEAST 48 HOURS PRIOR TO BEGINNING ANY EXCAVATION. THE CONTRACTOR SHALL GIVE ADEQUATE NOTIFICATION TO ALL AFFECTED UTILITY OWNERS FOR REMOVAL, RELOCATION AND ALTERATION OF THEIR EXISTING FACILITIES.
11. PROTECTION OF UTILITIES. THE CONTRACTOR SHALL BE RESPONSIBLE AT ALL TIMES THROUGHOUT THE DURATION OF CONSTRUCTION FOR THE PROTECTION OF ALL EXISTING AND NEWLY INSTALLED ABOVE GROUND, UNDERGROUND, AND ON THE SURFACE STRUCTURES AND UTILITIES FROM DAMAGE OR SERVICE DISRUPTION. THE CONTRACTOR SHALL BE RESPONSIBLE FOR TAKING SUCH MEASURES AS NECESSARY TO PROTECT THE HEALTH, SAFETY, AND WELFARE OF THOSE PERSONS HAVING ACCESS TO THE WORK SITE.
12. EXISTING UTILITIES. PRIOR TO BEGINNING CONSTRUCTION, THE CONTRACTOR SHALL VERIFY THE SIZE, LOCATION, ELEVATION, AND MATERIAL OF ALL EXISTING UTILITIES WITHIN THE AREA OF CONSTRUCTION. THE CONTRACTOR IS RESPONSIBLE FOR VERIFICATION OF ALL EXISTING UTILITIES AND SOILS CONDITIONS AND SHALL NOTIFY THE ENGINEER IN THE EVENT OF ANY DISCREPANCIES WITH THE PLANS.
13. EXISTING UTILITIES. THE CONTRACTOR SHALL BE RESPONSIBLE FOR DAMAGE TO ANY EXISTING UTILITIES FOR WHICH IT FAILS TO REQUEST LOCATIONS FROM THE UTILITY OWNER. THE CONTRACTOR IS RESPONSIBLE AS WELL FOR DAMAGE TO ANY EXISTING UTILITIES WHICH ARE PROPERLY LOCATED.
14. FIELD REVIEWS. THE CONTRACTOR SHALL NOTIFY THE ENGINEER 48 HOURS PRIOR TO ANY REQUIRED FIELD REVIEWS OR INSPECTIONS AND SHALL SUPPLY ALL EQUIPMENT NECESSARY, LABOR AND MATERIALS FOR INSPECTION AND/OR TEST. ALL WORK SHALL BE OPEN AND SUBJECT TO REVIEW AND/OR INSPECTION BY AUTHORIZED PERSONNEL OF THE CITY, OWNER, INVOLVED UTILITY COMPANIES, PROJECT ENGINEER, AND REGULATORY AGENCIES. ALL RECOMMENDATIONS AND REQUIREMENTS OF INSPECTION PERSONNEL OTHER THAN THE OWNER'S SHALL BE REPORTED TO THE ENGINEER PRIOR TO IMPLEMENTATION. COMPENSATION WILL NOT BE ALLOWED FOR WORK WHICH IS NOT AUTHORIZED BY THE ENGINEER OR OWNER.
15. DIFFERING SITE CONDITIONS. THE CONTRACTOR SHALL IMMEDIATELY NOTIFY THE ENGINEER OF ANY DISCREPANCIES FOUND BETWEEN THE DRAWINGS AND THE FIELD CONDITIONS PRIOR TO CONSTRUCTION IN THE AREA OF CONFLICT. ANY DIFFERING SITE CONDITIONS FROM THAT WHICH ARE REPRESENTED ON THE PLANS, WHETHER ABOVE, ON OR BELOW THE SURFACE OF THE GROUND, ARE TO BE BROUGHT TO THE IMMEDIATE ATTENTION OF THE ENGINEER AND THE OWNER IN WRITING.
16. DEWATERING. THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING AND COMPLYING WITH ANY DEWATERING PERMITS AND/OR APPROVALS NECESSARY FOR THE CONSTRUCTION OF THE PROPOSED FACILITIES AND SYSTEMS.
17. PERMITS. THE CONTRACTOR SHALL OBTAIN ALL REQUIRED PERMITS AND/OR LICENSES TO COMMENCE CONSTRUCTION.
18. USE OF EXISTING MATERIALS. NO EXISTING MATERIAL SHALL BE USED IN NEW CONSTRUCTION UNLESS APPROVED DURING THE SHOP DRAWING APPROVAL PROCESS.
19. EXISTING LANDSCAPE. CONTRACTOR SHALL BE EXTREMELY CAUTIOUS WHEN WORKING NEAR TREES WHICH ARE TO BE SAVED, WHETHER SHOWN ON THE PLAN OR DESIGNATED IN THE FIELD.
20. TESTING. THE CONTRACTOR SHALL BE RESPONSIBLE FOR ALL REQUIRED TESTS TO BE PERFORMED BY AN INDEPENDENT TESTING LABORATORY APPROVED BY THE ENGINEER. THE CONTRACTOR UNDERSTANDS THAT ANY AND ALL TESTS, INSPECTIONS, PROCEDURES INCLUDING ALL LABOR, MATERIALS, TIME AND EFFORT IN ORDER TO HAVE ALL WORK DESIGNATED AS FINAL BY THE RESPECTIVE JURISDICTIONAL AUTHORITIES AND THAT SUCH TESTS, INSPECTIONS AND PROCEDURES ARE INCLUDED IN THE BID.
21. FINAL UTILITY ADJUSTMENTS. ALL UTILITIES SHALL BE ADJUSTED TO NEW FINISH GRADES AND PROPERLY SET TO PAVEMENT CROSS SLOPE AS REQUIRED.
22. PERMITS. OWNER AND CONTRACTOR ARE RESPONSIBLE FOR TRACKING ALL PERMIT EXPIRATION DATES AND ADVISING ENGINEER IF THEY DESIRE TO APPLY FOR PERMIT EXTENSIONS OR RENEWALS PRIOR TO PERMIT EXPIRATION. OWNER AND CONTRACTOR ARE ADVISED THAT REGULATORY AGENCIES MAY DECLINE TO EXTEND OR RE-APPROVE APPROVALS AND/OR MAY REQUIRE DESIGN CHANGES, ADDITIONAL FEES, ADDITIONAL SURETY, RE-REVIEW, ETC. CONTRACTOR AND OWNER ARE RESPONSIBLE FOR ALL DELAYS, ADDITIONAL COSTS, CHANGE ORDERS, ETC. ASSOCIATED WITH FAILING TO COMPLETE WORK PRIOR TO PERMIT EXPIRATION.



PROJECT LOCATION MAP
N.T.S.



DRAWING INDEX

DWG. #	TITLE
C1.0	COVER SHEET
C2.1	PAVING, GRADING, & DRAINAGE PLAN
C2.2	GEOMETRIC, SIGNING, & STRIPING PLAN
C2.3	WATER & WASTEWATER PLAN
C3.1	PAVING & DRAINAGE DETAILS
C4.1-4.5	WATER & WASTEWATER DETAILS
C5.1	SITE DEMOLITION & EROSION CONTROL PLAN
C6.1-6.2	GENERAL NOTES & SPECIFICATIONS

GOVERNING SPECIFICATIONS
REFER TO PROJECT SPECIFICATION SHEET FOR DETAILED PROJECT AND MATERIAL REQUIREMENTS. ITEMS NOT EXPLICITLY ADDRESSED BY SPECIFICATION SHEETS SHALL BE CONSTRUCTED PER FDOT SPECIFICATIONS FOR ROAD & BRIDGE CONSTRUCTION (y2020-21) UNLESS OTHERWISE DIRECTED BY THE ENGIN[redacted]

LEGAL DESCRIPTION (FROM SURVEY):
BEGINNING AT THE NORTHEAST CORNER OF PARCEL III, ACCORDING TO THE PLAT OF 53RD STREET WAREHOUSES, AS RECORDED IN PLAT BOOK 67, PAGES 83 THROUGH 86, PUBLIC RECORDS OF PALM BEACH COUNTY, FLORIDA; THENCE SOUTH 01°33'25" WEST ALONG THE EAST LINE OF SAID PARCEL III A DISTANCE OF 269.45 FEET TO THE SOUTHEAST CORNER THEREOF; THENCE NORTH 88°26'35" WEST ALONG THE SOUTH LINE OF SAID PARCEL III A DISTANCE OF 193.75 FEET TO THE SOUTHWEST CORNER THEREOF; THENCE NORTH 01°33'25" EAST ALONG THE WEST LINE OF SAID PARCEL iii A DISTANCE OF 269.45 FEET TO THE NORTHWEST CORNER THEREOF; THENCE SOUTH 88°26'35" EAST ALONG THE NORTH LINE OF SAID PARCEL III A DISTANCE OF 193.75 FEET TO THE POINT OF BEGINNING.

Know what's **below.**
Call before you dig.

ALL ELEVATIONS REFER TO NORTH AMERICAN VERTICAL DATUM OF 1988 (NAVD 88). TO CONVERT TO NGVD 29 DATUM, ADD 1.54' TO NAVD ELEVATIONS.

NOTE: THESE PLANS ARE PRELIMINARY AND SHOULD NOT BE USED FOR CONSTRUCTION OR BIDDING/CONTRACTUAL PURPOSES UNTIL ALL PERMITS ARE ISSUED.

PROJECT TEAM

[redacted]
1319 MANGONIA PARK, LLC

CIVIL E[redacted]
McLEOD, McCARTHY & ASSOCIATES
TODD N. McLEOD, P.E.
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PROFESSIONAL ENGINEER
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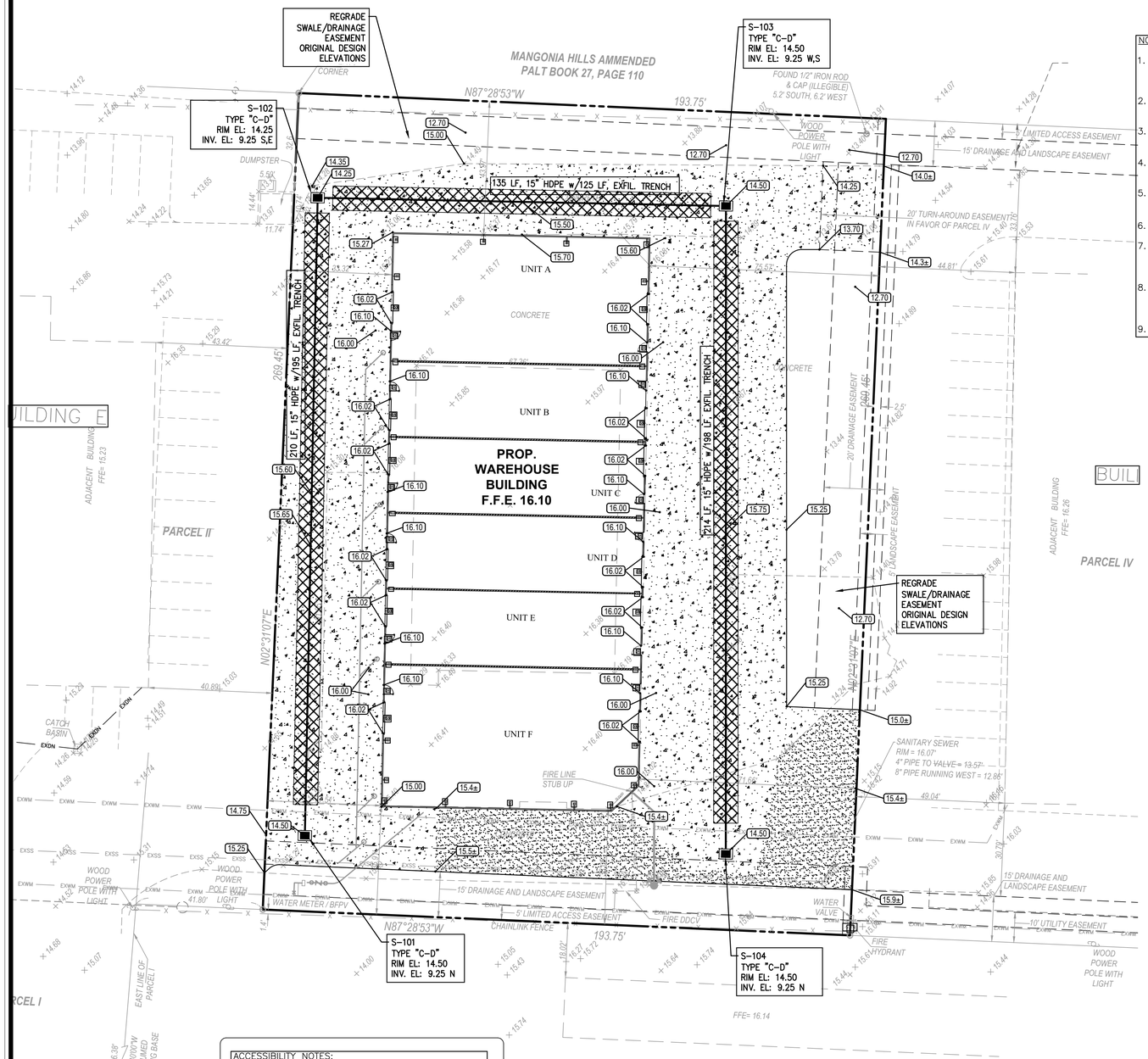
FIELD:
DRAWN: P. Saffold
DESIGNED: TNM
APPROVED: TNM
PROJECT #21-030

NO.	DATE	REVISIONS
1	1/25/2022	REV PER TOWN

COVER SHEET
1319 MANGONIA PARK WAREHOUSE
MANGONIA PARK, FLORIDA

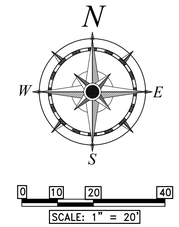
TOWN RESUBMITTAL
DATE: 8/1/2024

SHEET
C1.0
OF 13



NOTES:

1. EXIST. UTILITIES, DRAINAGE, & ELEVATIONS BASED ON SURVEY PREPARED BY PM SURVEYING. CONTRACTOR SHALL VERIFY INVERTS, PIPE SIZES, AND STRUCTURE LOCATIONS PRIOR TO SUBMITTING SHOP DRAWINGS.
2. ALL LANDSCAPE AREAS ADJACENT TO BUILDING SHALL BE GRADED TO DRAIN AWAY FROM BUILDING.
3. CONTRACTOR SHALL CONFIRM DETECTABLE WARNING REQUIREMENTS WITH BUILDING OFFICIAL PRIOR TO INSTALLATION.
4. WHERE LANDSCAPE/SOD ABUTS BUILDING SLAB, A MINIMUM 3" SLAB REVEAL SHALL BE PROVIDED.
5. MAXIMUM SLOPE FROM TOP OF CURB AND BACK OF WALK TO FINISHED GRADE SHALL BE 4(H):1(V), UNLESS OTHERWISE NOTED.
6. ALL BUFFER & UNPAVED/UNLANDSCAPED AREAS SHALL BE SODDED BY CONTRACTOR.
7. ALL OFFSITE DISTURBED AREAS SHALL BE SODDED BY CONTRACTOR. ALL DAMAGED CURBING, PAVEMENT, STRIPING, SIGNAGE, LANDSCAPING, ETC. SHALL BE RESTORED BY CONTRACTOR.
8. CONTRACTOR IS SOLELY RESPONSIBLE FOR PREPARING MAINTENANCE OF TRAFFIC (MOT) PLANS AND OBTAINING ALL REGULATORY APPROVALS FOR MOT PLANS. COPIES OF APPROVED MOT PLANS SHALL BE PROVIDED TO ENGINEER PRIOR TO COMMENCING WORK.
9. ALL ELEVATIONS REFER TO NORTH AMERICAN VERTICAL DATUM OF 1988 (NAVD 88).



LEGEND

- PROP. SIGN
- SURFACE FLOW ARROW
- PROP. CONCRETE PAVEMENT
- EXIST. CONC. PAVEMENT
- MILLING & RESURFACING (1" SP-9.5)
- PROPOSED ELEVATION (XX.XX)
- EXIST. ELEVATION (+16.17)

DRAINAGE STRUCTURE CHART

STRUCT. NUMBER	STRUCT. TYPE		GRATE TYPE
S-101	C		U.S.F. 4139-6429
S-102	C		U.S.F. 4139-6429
S-103	C		U.S.F. 4139-6429
S-104	C		U.S.F. 4139-6429

SG = SEATED GRATE

ACCESSIBILITY NOTES:

1. ACCESSIBLE ROUTES SHALL BE CONSTRUCTED TO MEET THE REQUIREMENTS OF THE FLORIDA BUILDING CODE – ACCESSIBILITY.
2. ALL WALKS CROSSING A VEHICULAR AREA SHALL HAVE DETECTABLE WARNING SURFACE (TRUNCATED DOME) IN ACCORDANCE WITH THE FLORIDA BUILDING CODE – ACCESSIBILITY.
3. CURB RAMP SLOPES AND DIMENSIONS SHALL BE IN ACCORDANCE WITH THE FLORIDA DEPARTMENT OF TRANSPORTATION (FDOT) STANDARD INDEX NO. 522-02. CURB RAMP DETECTABLE WARNING SURFACES SHALL BE TRUNCATED DOME AND SHALL BE IN ACCORDANCE WITH THE FLORIDA BUILDING CODE.
4. DETECTABLE WARNING SHALL BE THE WIDTH OF THE WALKING SURFACE AND 3' IN LONGITUDINAL LENGTH (DIRECTION OF TRAVEL). DETECTABLE WARNING MATS SHALL PER THE FDOT APPROVED PRODUCTS LIST (APL), LATEST EDITION.

ALL ELEVATIONS REFER TO NORTH AMERICAN VERTICAL DATUM OF 1988 (NAVD 88). TO CONVERT TO NGVD 29 DATUM, ADD 1.54' TO NAVD ELEVATIONS.

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FIELD:
DRAWN: P. Saffold
DESIGNED: TNM
APPROVED: TNM

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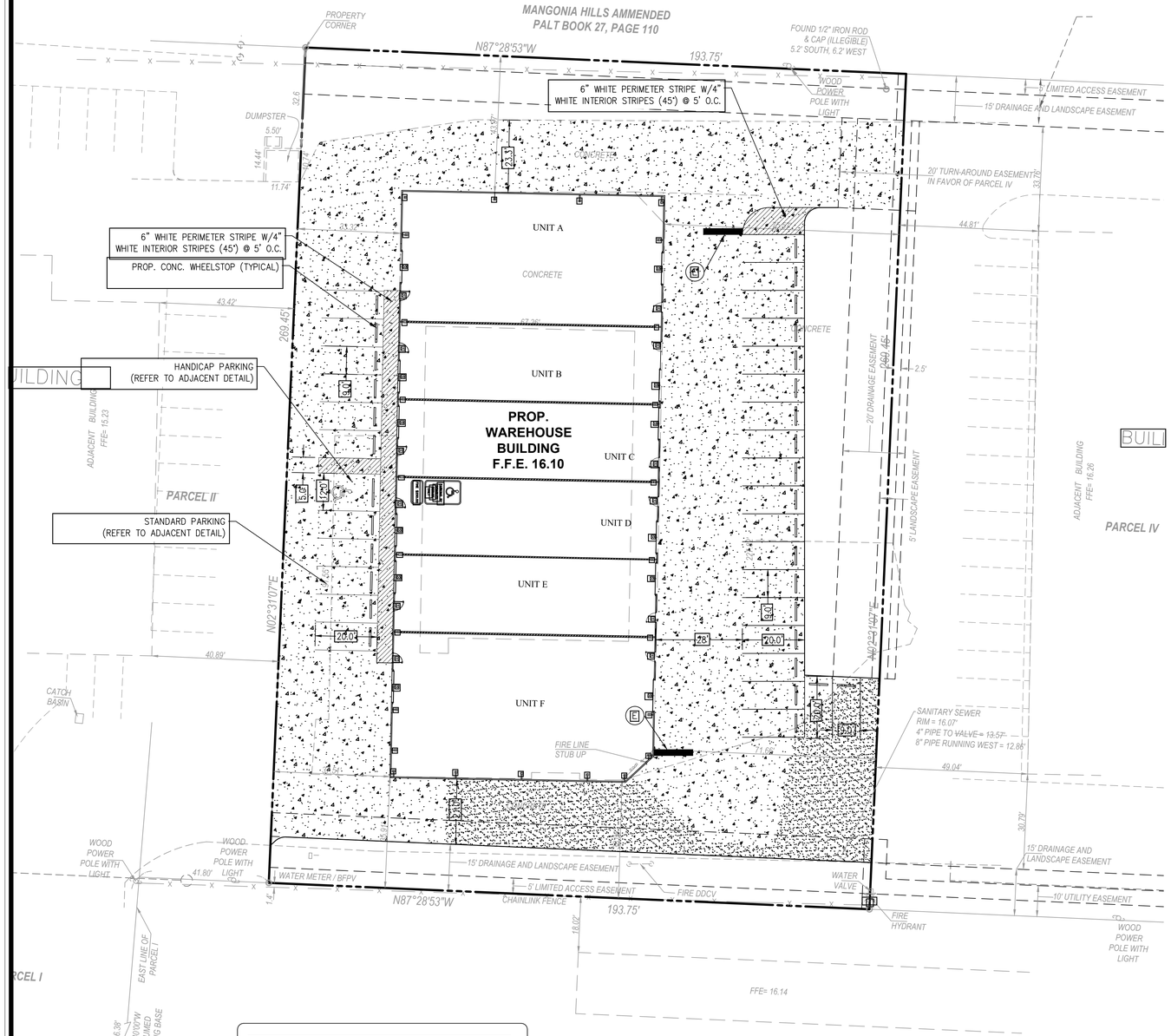
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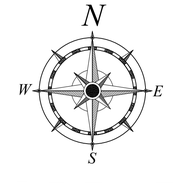
PAVING, GRADING, & DRAINAGE PLAN
1319 MANGONIA PARK WAREHOUSE
MANGONIA PARK, FLORIDA

TOWN RESUBMITTAL
DATE: 8/1/2024

SHEET
C2.1
OF 13



STRIPING LEGEND	
6" SOLID WHITE STRIPE	A
8" SOLID WHITE STRIPE	B
12" SOLID WHITE STRIPE	C
18" SOLID WHITE STRIPE	D
24" SOLID WHITE STRIPE	E
6" SKIP WHITE (10'–30')	F
6" SKIP WHITE (6'–10')	G
6" SKIP WHITE (2'–4')	H
6" SOLID YELLOW	I
18" SOLID YELLOW	J
6" DOUBLE YELLOW	K
6" SKIP YELLOW (10'–30')	L
6" SKIP YELLOW (6'–10')	M
6" SKIP YELLOW (2'–4')	N
RPM BI–DIRECTIONAL AMBER/AMBER	P
FDP WHITE	R
FDP YELLOW	S
RPM BI–DIRECTIONAL WHITE/RED	T
RPM BI–DIRECTIONAL RED/YELLOW	U



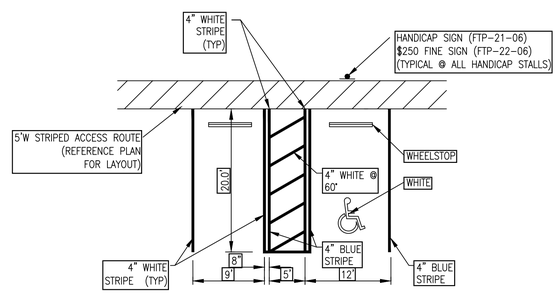
LEGEND	
PROP. SIGN	
CURB RAMP PER FDOT INDEX NO. 522–002	CR–X
PROP. CONCRETE PAVEMENT	
EXIST. CONC. PAVEMENT	
MILLING & RESURFACING	

CURBING LEGEND	
TYPE "D" CURB	1
TYPE "F" CURB & GUTTER	2
2' VALLEY GUTTER	3
12" HEADER CURB	4
DETECTABLE WARNING (PER FDOT 522–02) TYPE "B" CURB	5

NOTES:
1. PAVEMENT MARKINGS & SIGNAGE SHALL BE IN ACCORDANCE WITH THE MANUAL ON UNIFORM TRAFFIC CONTROL DEVICES FOR STREETS AND HIGHWAYS (MUTCD) AND TOWN OF MANGONIA PARK LAND DEVELOPMENT CODE.
2. PARKING STALL STRIPING SHALL BE TRAFFIC PAINT; ALL OTHER STRIPING SHALL BE THERMOPLASTIC.
3. CURB RAMPS SHALL BE CONSTRUCTED IN ACCORDANCE WITH FDOT STANDARD INDEX FOR ROAD CONSTRUCTION NO. 522–002.

ACCESSIBILITY NOTES:

1. ACCESSIBLE ROUTES SHALL BE CONSTRUCTED TO MEET THE REQUIREMENTS OF THE FLORIDA BUILDING CODE – ACCESSIBILITY.
2. ALL WALKS CROSSING A VEHICULAR AREA SHALL HAVE DETECTABLE WARNING SURFACE (TRUNCATED DOME) IN ACCORDANCE WITH THE FLORIDA BUILDING CODE – ACCESSIBILITY.
3. CURB RAMP SLOPES AND DIMENSIONS SHALL BE IN ACCORDANCE WITH THE FLORIDA DEPARTMENT OF TRANSPORTATION (FDOT) STANDARD INDEX NO. 522–02. CURB RAMP DETECTABLE WARNING SURFACES SHALL BE TRUNCATED DOME AND SHALL BE IN ACCORDANCE WITH THE FLORIDA BUILDING CODE.
4. DETECTABLE WARNING SHALL BE THE WIDTH OF THE WALKING SURFACE AND 3' IN LONGITUDINAL LENGTH (DIRECTION OF TRAVEL). DETECTABLE WARNING MATS SHALL PER THE FDOT APPROVED PRODUCTS LIST (APL), LATEST EDITION.



TYPICAL 90° PARKING STALL AND HANDICAP SPACE DETAIL
NOTE: HANDICAP STALL SHALL HAVE MAXIMUM 2.0% SLOPE IN ANY DIRECTION N.T.S.

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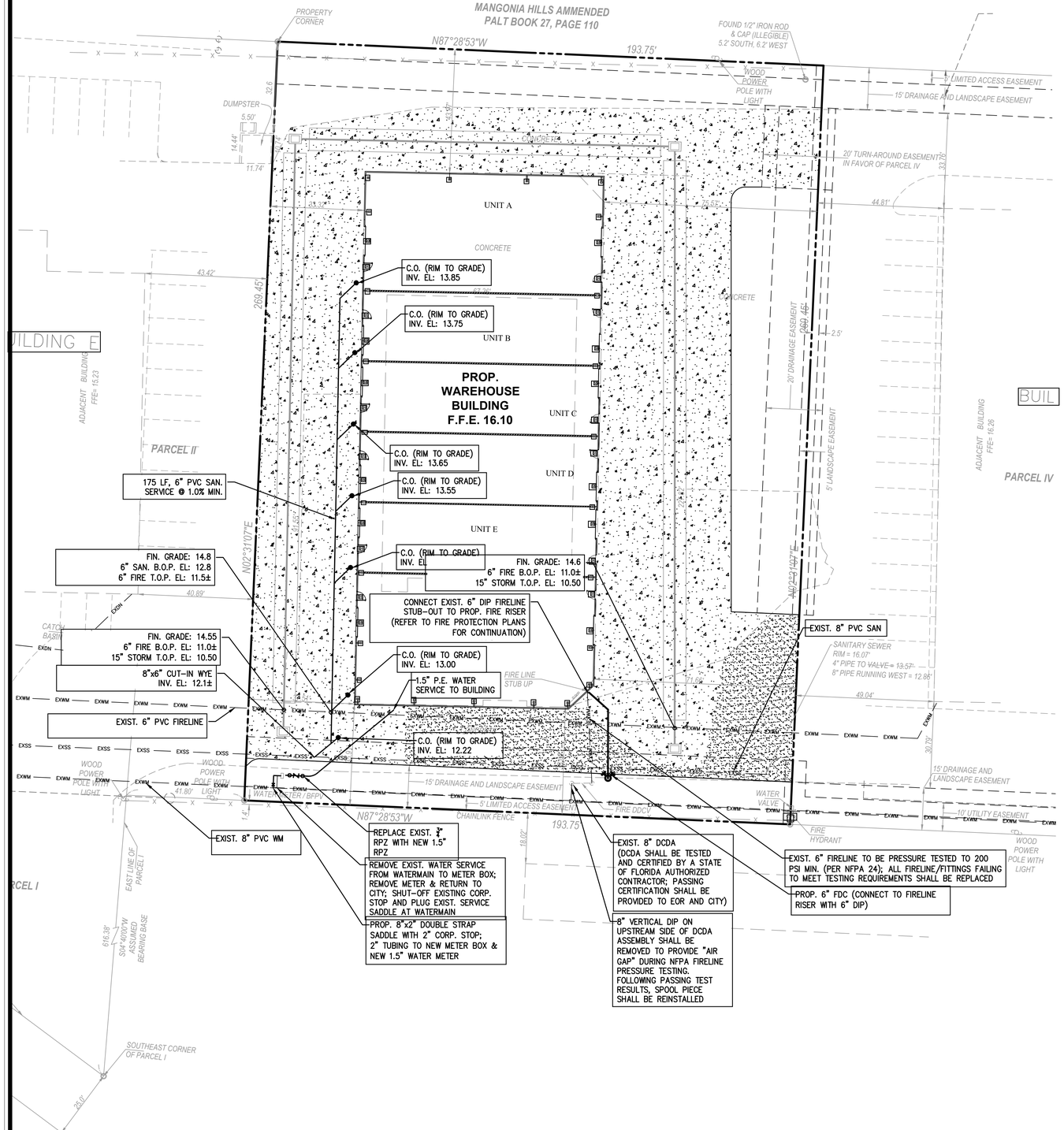
PROJECT #21-030

NO.	DATE	REVISIONS

GEOMETRIC, SIGNING, & MARKING PLAN
1319 MANGONIA PARK WAREHOUSE
MANGONIA PARK, FLORIDA

TOWN RESUBMITTAL
DATE: 8/1/2024

SHEET
C2.2
OF 13



LEGEND

- EASEMENT LINE
- EXISTING WATER MAIN, GATE VALVE (G.V.), TEE, F.H. AND PLUG
- EXISTING SANITARY SEWER MAIN, MANHOLE AND DIRECTION OF FLOW
- PROPOSED WATER MAIN, GATE VALVE (G.V.), BEND AND TEE WITH FIRE HYDRANT ASSEMBLY
- PROPOSED SANITARY SEWER MAIN, SINGLE & DOUBLE SEWER SERVICE, CLEAN OUT (C.O.), MANHOLE AND
- CONCRETE
- MILLING & RESURFACING (1" SP-9.5)
- EXIST. CONCRETE
- UTILITY CONFLICT
- U.E. UTILITY EASEMENT

N

W E

S

0 10 20 40

SCALE: 1" = 20'

RECORD DRAWING NOTES:
1. RECORD DRAWINGS SHALL BE PREPARED IN THE STATE PLANE COORDINATE SYSTEM.
2. ALL UTILITY FEATURES SHALL BE SHOWN IN THEIR AS-BUILT LOCATION.
3. STATE PLANE COORDINATES SHALL BE DISPLAYED ON RECORD DRAWINGS FOR ALL FEATURES SPECIFIED IN THE TOWN OF MANGONIA PARK UTILITY STANDARDS.
4. STATE PLANE COORDINATES SHALL BE SHOWN ON PROPERTY CORNERS.

GENERAL WATER NOTES:
1. ALL WATER MAIN DUCTILE IRON PIPE (DIP) AND PIPE FITTINGS SHALL BE PAINTED WITH A 4" WIDE CONTINUOUS BLUE LINE THAT RUNS PARALLEL TO THE AXIS OF THE PIPE AND IS LOCATED ALONG THE TOP OF THE PIPE.
2. ALL WATER MAINS SHALL BE MARKED WITH ONE CONTINUOUS STRIP OF 6" WIDE MAGNETIC BLUE CODED TAPE IMPRINTED WITH TWO (2) INCH HIGH LETTERING READING "CAUTION – POTABLE WATER LINE BURIED BELOW" AND LOCATED APPROXIMATELY TWELVE (12) INCHES ABOVE THE CROWN OF THE PIPE. THE WORDING SHALL OCCUR EVERY THREE (3) FEET.
3. ALL WATER SERVICE BRASS ARE REQUIRED TO BE LEAD FREE.
4. ALL EXISTING VALVE BOXES WITHIN PROJECT LIMITS SHALL BE REPLACED. BOXES SHALL BE ADJUSTED TO FINAL FINISHED GRADE, AS REQUIRED. VALVE BOXES IN GREEN/UNPAVED AREAS SHALL HAVE 2'x2' CONC. COLLARS POURED.
5. ALL BACKFLOW PREVENTION DEVICES SHALL BE USC-APPROVED.

GENERAL SEWER NOTES:
1. ON-SITE SEWER LATERALS ARE PRIVATELY OWNED AND MAINTAINED.

NOTES:
1. PRIOR TO CONSTRUCTION, IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO FIELD VERIFY (ELEVATION & LOCATION) ALL EXISTING UTILITIES. ANY DISCREPANCIES SHALL BE BROUGHT TO THE ATTENTION OF THE ENGINEER PRIOR TO CONTINUATION OF WORK.
2. ALL SANITARY SEWER SHALL BE P.V.C. SDR 26 (UNLESS OTHERWISE NOTED).
3. ALL WATER MAINS & FORCE MAINS SHALL BE DUCTILE IRON UNLESS OTHERWISE NOTED.
4. FIRE HYDRANTS SHALL BE LOCATED A MINIMUM OF 7.5' AND A MAXIMUM OF 12' FROM EDGE OF PAVEMENT.
5. VALVES SHALL NOT BE PLACED IN CURBS.
6. ALL WATER AND SANITARY SERVICES SHALL BE CONSTRUCTED A MINIMUM OF 5' FROM DRAINAGE STRUCTURES.
7. MAGNETIC TAPE IS REQUIRED FOR ALL WATER & FORCE MAINS AND SHALL BE 6" WIDE, IMPRINTED WITH 2" HIGH LETTERING READING "CAUTION – WATER MAIN BURIED BELOW" OR "CAUTION – FORCE MAIN BURIED BELOW", AS REQUIRED.
8. RECORD DRAWINGS MUST HAVE DATA FOR TAP AND METER LOCATIONS FOR ALL NON-PERPENDICULAR WATER MAIN SERVICES.
9. PRESSURE TESTING AND CERTIFICATION SHALL FOLLOW TOWN OF MANGONIA PARK UTILITY CONSTRUCTION STANDARDS, LATEST EDITION.
10. ALL WATERMAINS TO CROSS OVER SANITARY SEWER SERVICES. WHERE 18" MINIMUM CLEARANCE CANNOT BE MAINTAINED, 20 LINEAR FEET OF DUCTILE IRON WATERMAIN SHALL BE CENTERED ON THE CROSSING AND THE SEWER SERVICE SHALL BE C-900, SDR-18. IN NO CASE SHALL THERE BE LESS THAN 6" CLEARANCE.
13. WATER/SEWER SYMBOLS ARE REPRESENTATIVE OF THE CORRESPONDING FITTINGS ONLY. CONTRACTOR SHALL NOT ATTEMPT TO ATTEMPT TO CONSTRUCT WATER/SEWER SYSTEM BY SCALING THE SYMBOLS FORM THESE PLANS. ALL FITTINGS SHALL INSTEAD BE BUILT PER THE ENCLOSED DETAILS.
14. ALL PRIVATE FIRE LINES AND APPURTENANCES SHALL BE CONSTRUCTED AND TESTED IN ACCORDANCE WITH NFPA STANDARDS.

WELLFIELD PROTECTION NOTES:
PORTIONS OF THE EXISTING SANITARY SEWER SYSTEM ARE WITHIN ZONES 1 AND 2 OF A TOWN WELLFIELD PROTECTION ZONE. CONTRACTOR SHALL CONFORM WITH THE PBC WELLFIELD PROTECTION ORDINANCE AND SHALL TEST NEWLY INSTALLED SANITARY SERVICES AS FOLLOWS (ULDC ARTICLE 14, CHAPTER "B", WELLFIELD PROTECTION):

B. POLYVINYL CHLORIDE PIPE (PVC) (GASKETED JOINT) AND FITTINGS FOR GRAVITY WASTEWATER AND SEWER FORCE MAIN APPLICATIONS
1. PIPE FOUR INCHES OR LARGER IN DIAMETER SHALL CONFORM TO THE REQUIREMENTS AS SET FORTH IN AWWA C900/C905 WITH MINIMUM DIMENSION RATIO DR 18. PROVISIONS MUST BE MADE FOR CONTRACTION AND EXPANSION AT EACH JOINT, OR WITH A RUBBER RING AND AN INTEGRAL BELL AS PART OF EACH JOINT, OR BY A RUBBER RING-SEALED COUPLING. CLEAN, REWORKED MATERIAL GENERATED FROM THE MANUFACTURER'S OWN PIPE PRODUCTION MAY BE USED. FITTINGS SHALL BE CAST OR DUCTILE IRON. PIPE SHALL HAVE CAST IRON PIPE-EQUIVALENT OUTSIDE DIMENSIONS. [ORD. 2013-001]

2. PIPE SMALLER THAN FOUR INCHES IN DIAMETER SHALL CONFORM TO COMMERCIAL STANDARD CS 256 AND ASTM D22141. PROVISIONS SHALL BE MADE FOR CONTRACTION AND EXPANSION AT EACH JOINT WITH A RUBBER RING, AND AN INTEGRAL BELL AS PART OF EACH JOINT, OR BY A RUBBER RING-SEALED COUPLING. PIPE SHALL BE MADE FROM SDR 21, 200 PSI CLEAN, VIRGIN NSF-APPROVED TYPE I, GRADE 1 PVC CONFORMING TO ASTM D1784. CLEAN, REWORKED MATERIAL GENERATED FROM THE MANUFACTURER'S OWN PIPE PRODUCTION MAY BE USED. FITTINGS FOR PIPE SMALLER THAN FOUR INCHES IN DIAMETER SHALL BE PVC. [ORD. 2013-001]

NOTE: ALL WATER MAIN PIPE, INCLUDING FITTINGS, INSTALLED ON OR AFTER AUGUST 28, 2003, EXCEPT PIPE INSTALLED UNDER CONSTRUCTION PERMIT FOR WHICH THE DEPARTMENT RECEIVED A COMPLETE APPLICATION BEFORE AUGUST 28, 2003, SHALL BE COLOR CODED OR MARKED USING BLUE AS A PREDOMINANT COLOR TO DIFFERENTIATE DRINKING WATER FROM RECLAIMED OR OTHER WATER. UNDERGROUND PLASTIC PIPE SHALL BE SOLID-WALL BLUE PIPE, SHALL HAVE A CO-EXTRUDED BLUE EXTERNAL SKIN, OR SHALL BE WHITE OR BLACK WITH BLUE STRIPES INCORPORATED INTO, OR APPLIED TO, THE PIPE WALL; AND UNDERGROUND METAL OR CONCRETE PIPE SHALL HAVE BLUE STRIPES APPLIED TO THE PIPE WALL. PIPE STRIPED DURING MANUFACTURING OF THE PIPE SHALL HAVE CONTINUOUS STRIPES THAT RUN PARALLEL TO THE AXIS OF THE PIPE, THAT ARE LOCATED AT NO GREATER THAN 90-DEGREE INTERVALS AROUND THE PIPE, AND THAT WILL REMAIN INTACT DURING AND AFTER INSTALLATION OF THE PIPE. IF TAPE OR PAINT IS USED TO STRIPE PIPE DURING INSTALLATION OF THE PIPE, THE TAPE OR PAINT SHALL BE APPLIED IN A CONTINUOUS LINE THAT RUNS PARALLEL TO THE AXIS OF THE PIPE AND THAT IS LOCATED ALONG THE TOP OF PIPE; FOR PIPES WITH AN INTERNAL DIAMETER OF 24 INCHES OR GREATER, TAPE OR PAINT SHALL BE APPLIED IN CONTINUOUS LINES ALONG EACH SIDE OF THE PIPE AS WELL AS ALONG THE TOP OF THE PIPE. ABOVEGROUND PIPE AT DRINKING WATER TREATMENT PLANTS SHALL BE COLOR CODED AND LABELED IN ACCORDANCE WITH SUBSECTION 62-555.320(10), F.A.C., AND ALL OTHER ABOVEGROUND PIPE SHALL BE PAINTED BLUE OR COLOR CODED OR MARKED LIKE UNDERGROUND PIPE.

ALL ELEVATIONS REFER TO NORTH AMERICAN VERTICAL DATUM OF 1988 (NAVD 88). TO CONVERT TO NGVD 29 DATUM, ADD 1.54' TO NAVD ELEVATIONS.

NOTE: THESE PLANS ARE PRELIMINARY AND SHOULD NOT BE USED FOR CONSTRUCTION OR BIDDING/CONTRACTUAL PURPOSES UNTIL ALL PERMITS ARE ISSUED.

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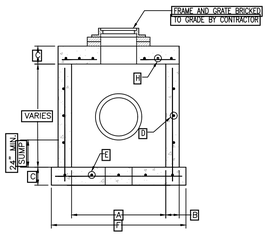
FIELD:
DRAWN: P. Saffold
DESIGNED: TNM
APPROVED: TNM
PROJECT #21-030

NO.	DATE	REVISIONS
1	1/23/2022	REV PER TOWN

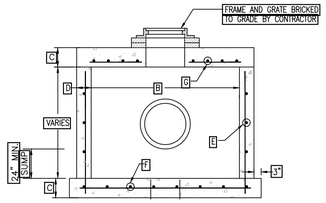
WATER & WASTWATER PLAN
1319 MANGONIA PARK WAREHOUSE
MANGONIA PARK, FLORIDA

TOWN RESUBMITTAL
DATE: 8/1/2024

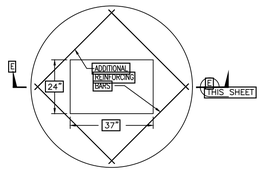
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OF 13



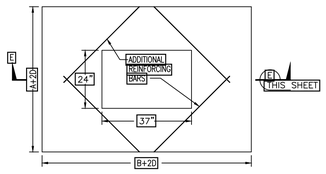
CATCH BASIN
(SEE TABLE FOR DIMENSIONS)



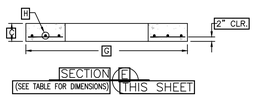
CATCH BASIN
(SEE TABLE FOR DIMENSIONS)



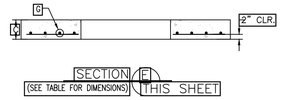
TOP SLAB FOR CATCH BASIN



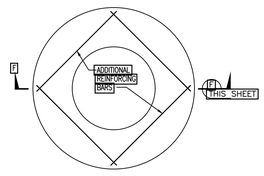
TOP SLAB FOR CATCH BASIN



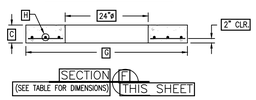
SECTION (E) THIS SHEET
(SEE TABLE FOR DIMENSIONS)



SECTION (E) THIS SHEET
(SEE TABLE FOR DIMENSIONS)



TOP SLAB FOR MANHOLE



SECTION (E) THIS SHEET
(SEE TABLE FOR DIMENSIONS)

A	B	C	D	E*	F*	G*
3'-6"	3'-6"	8"	6"	#4 @ 12"	#4 @ 9"	#4 @ 6"
4'-0"	4'-0"	8"	6"	#4 @ 12"	#4 @ 9"	#4 @ 6"
4'-10"	5'-0"	8"	8"	#4 @ 12"	#5 @ 12"	#5 @ 6"
6'-0"	6'-0"	8"	8"	#4 @ 12"	#6 @ 12"	#6 @ 6"
8'-0"	8'-0"	10"	8"	#4 @ 12"	#6 @ 12"	#6 @ 6"
10'-0"	10'-0"	10"	8"	#4 @ 12"	#6 @ 6"	#7 @ 6"
3'-0"	4'-6"	8"	8"	#4 @ 12"	#4 @ 12"	#4 @ 6"
3'-6"	6'-0"	8"	8"	#4 @ 12"	#5 @ 12"	#5 @ 6"
4'-0"	6'-0"	8"	8"	#4 @ 12"	#5 @ 12"	#5 @ 6"
3'-6"	8'-0"	8"	8"	#4 @ 12"	#5 @ 12"	#5 @ 6"
4'-0"	8'-0"	8"	8"	#4 @ 12"	#5 @ 12"	#5 @ 6"
5'-0"	7'-0"	8"	8"	#4 @ 12"	#5 @ 12"	#5 @ 6"
6'-0"	8'-0"	8"	8"	#4 @ 12"	#5 @ 12"	#5 @ 6"
7'-0"	9'-0"	8"	8"	#4 @ 12"	#5 @ 12"	#5 @ 6"
8'-0"	10'-0"	10"	8"	#4 @ 12"	#6 @ 12"	#6 @ 6"
10'-0"	12'-0"	12"	8"	#4 @ 12"	#6 @ 12"	#6 @ 6"

INLET NOTES:

BEVELED EDGES: ALL EXPOSED CORNERS AND EDGES TO BE CHAMFERED 3/4".
FOUNDATION MATERIAL: WHERE MATERIAL UNSATISFACTORY FOR FOUNDATION IS ENCOUNTERED, ALL SUCH MATERIAL MUST BE REMOVED DOWN TO SATISFACTORY MATERIAL AND BACKFILLED TO SUBGRADE WITH CLEAN SAND.
INLET TYPES: INLETS ARE TO BE CONSTRUCTED TO THE DIMENSIONS SHOWN HEREON. INLETS RECEIVING PIPE LARGER THAN 42" DIAMETER SHALL BE IN ACCORDANCE WITH F.D.O.T. STANDARDS.
MATERIAL: INLET WALLS AND BASES MAY EITHER BE CAST-IN-PLACE CLASS I, 2500 P.S.I. CONCRETE OR PRECAST CLASS II, 4000 (MIN.) P.S.I. CONCRETE.

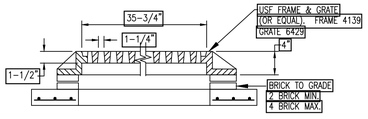
RECTANGULAR CATCH BASIN (ON-SITE)

A	B	C	D	E* C.C.E.W.	F	G	H* C.C.E.W.
3'-6"	4"	8"	ASTM C-478	#4 @ 12"	4'-8"	4'-2"	#4 @ 6"
3'-6"	6"	8"	ASTM C-478	#4 @ 12"	5'-0"	4'-6"	#4 @ 6"
4'-6"	6"	8"	ASTM C-478	#4 @ 12"	6'-0"	5'-0"	#4 @ 6"
4'-6"	8"	8"	ASTM C-478	#4 @ 12"	6'-4"	5'-4"	#4 @ 6"
5'-0"	8"	8"	ASTM C-478	#5 @ 12"	7'-4"	6'-4"	#5 @ 8"
6'-0"	8"	8"	ASTM C-478	#5 @ 6"	8'-0"	7'-0"	#5 @ 6"
6'-0"	8"	8"	ASTM C-478	#5 @ 6"	8'-4"	7'-4"	#5 @ 6"
7'-0"	8"	8"	ASTM C-478	#5 @ 6"	9'-4"	8'-4"	#5 @ 6"
8'-0"	10"	10"	ASTM C-478	#5 @ 6"	10'-8"	9'-8"	#6 @ 6"
10'-0"	12"	12"	ASTM C-478	#5 @ 6"	12'-0" SQ	12'-0"	#6 @ 6"

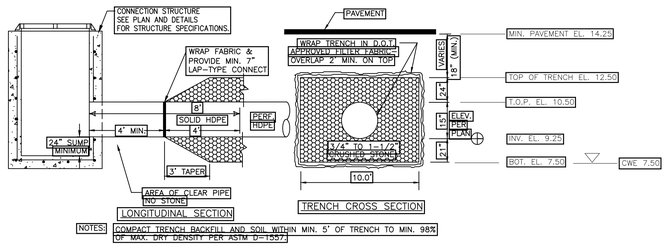
INLET NOTES:

BEVELED EDGES: ALL EXPOSED CORNERS AND EDGES TO BE CHAMFERED 3/4".
FOUNDATION MATERIAL: WHERE MATERIAL UNSATISFACTORY FOR FOUNDATION IS ENCOUNTERED, ALL SUCH MATERIAL MUST BE REMOVED DOWN TO SATISFACTORY MATERIAL AND BACKFILLED TO SUBGRADE WITH CLEAN SAND.
INLET TYPES: INLETS ARE TO BE CONSTRUCTED TO THE DIMENSIONS SHOWN HEREON. INLETS RECEIVING PIPE LARGER THAN 42" DIAMETER SHALL BE IN ACCORDANCE WITH F.D.O.T. STANDARDS.
MATERIAL: INLET WALLS AND BASES MAY EITHER BE CAST-IN-PLACE CLASS I, 2500 P.S.I. CONCRETE OR PRECAST CLASS II, 4000 (MIN.) P.S.I. CONCRETE.

CIRCULAR CATCH BASIN (ON-SITE)

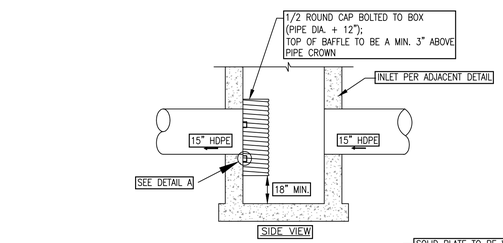


ON-SITE INLET FRAME & GRATE
(TYPE "C-D" INLET)



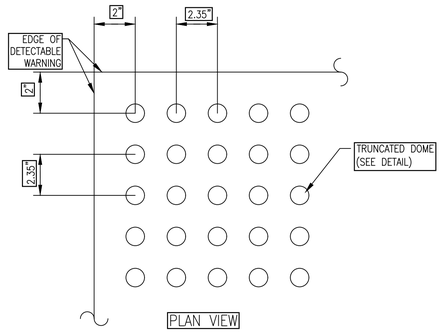
NOTES: COMPACT TRENCH BACKFILL AND SOIL WITHIN MIN. 5' OF TRENCH TO MIN. 98% OF MAX. DRY DENSITY PER ASTM D-1557.

ON-SITE EXFILTRATION TRENCH DETAIL
N.T.S.

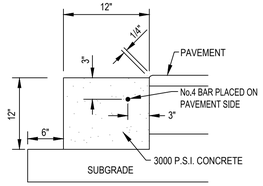


NOTE: BAFFLE TO BE USED ON ALL PIPE INVERTS CONNECTING TO EXFILTRATION TRENCH.

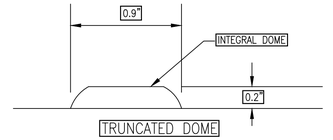
POLLUTION RETARDANT BAFFLE (PRB) DETAIL
N.T.S.



PLAN VIEW

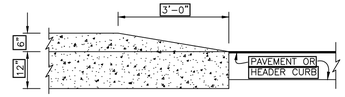


12" FLUSH HEADER CURB
N.T.S.



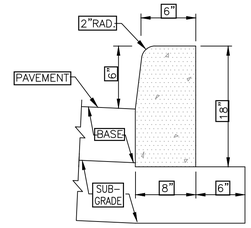
TRUNCATED DOME

TYPICAL CURB RAMP DETECTABLE WARNING DETAIL
(COMPLY WITH FDOT INDEX 304 & FBC)
N.T.S.

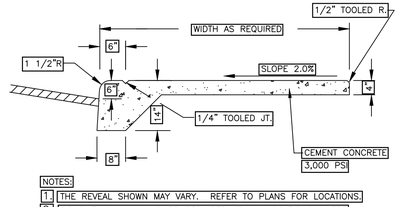


NOTE:
WHEN CURB TRANSITIONS ARE ADJACENT TO SIDEWALK THE TRANSITION MUST BE 12:1

TRANSITION CURB TAPER DETAIL
N.T.S.



TYPE "D" CURB
N.T.S.



NOTES:
1. THE REVEAL SHOWN MAY VARY. REFER TO PLANS FOR LOCATIONS.
2. MONOLITHIC WALK & CURB REQUIRED AT ALL LOCATIONS WHERE PROPOSED SIDEWALK ABUTS ASPHALT PAVEMENT.

MONOLITHIC CURB AND SIDEWALK SECTION
N.T.S.

PAVEMENT SECTION REQUIREMENTS

TYPE	WEARING SURFACE	BASE	SUB-GRADE
CONCRETE PAVEMENT (ON-SITE)	6" THICK (4,000 PSI) WITH JOINTING PER ACI 330. JOINTING PLAN TO BE SUBMITTED PRIOR TO POURING CONCRETE	N/A	12" THICK, COMPACTED TO 98% MAXIMUM DRY DENSITY A.A.S.H.T.O. T-180 (LBR 30)
ASPHALT PAVEMENT (ON-SITE)	2.0" THICK SP-9.5 A.C.S.C. (2-LIFTS)	8" THICK LIMEROCK BASE (LBR 100) COMPACTED TO 98% AASHTO T-180 DENSITY & PLACED IN 4" MAX. LIFTS.	12" THICK, COMPACTED TO 98% MAXIMUM DRY DENSITY A.A.S.H.T.O. T-180
SIDEWALKS	SIDEWALK: 4" THICK CONCRETE (3,000 PSI). CONCRETE TO BE BROOM FINISHED WITH EVEN, DUSTLESS SURFACE. CONTROL JOINTS TO BE SPACED 5' ON CENTER. ISOLATION JOINTS REQUIRED WHERE SIDEWALK ABUTS BUILDING, EXISTING PAVEMENT, OR OTHER STRUCTURES.		

ALL ELEVATIONS REFER TO NORTH AMERICAN VERTICAL DATUM OF 1988 (NAVD 88). TO CONVERT TO NGVD 29 DATUM, ADD 1.54' TO NAVD ELEVATIONS.

NOTE: THESE PLANS ARE PRELIMINARY AND SHOULD NOT BE USED FOR CONSTRUCTION OR BIDDING/CONTRACTUAL PURPOSES UNTIL ALL PERMITS ARE ISSUED.

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FIELD:
DRAWN: P. Saffold
DESIGNED: TNM
APPROVED: TNM

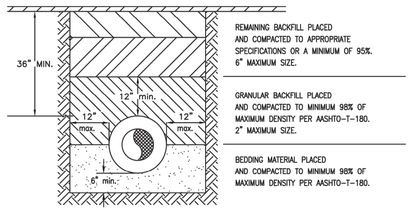
PROJECT #21-030

NO.	DATE	REVISIONS
1	1/25/2022	REV PER TOWN

PAVING & DRAINAGE DETAILS
1319 MANGONIA PARK WAREHOUSE
MANGONIA PARK, FLORIDA

TOWN RESUBMITTAL
DATE: 8/1/2024

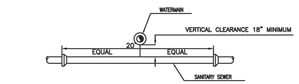
SHEET
C3.1
OF 13



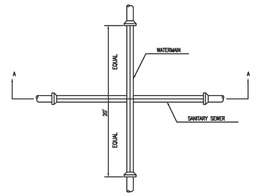
NOTES:

1. BEDDING SHALL CONSIST OF IN-SITU GRANULAR MATERIAL OR WASHED AND GRADED LIMEROCK 3/8"-7/5" SIZING.
2. THE PIPING SHALL BE FULLY SUPPORTED FOR ITS ENTIRE LENGTH WITH APPROPRIATE COMPACTION UNDER THE PIPE HAUNCHES.
3. THE PIPE SHALL BE PLACED IN A DRY TRENCH.
4. BACKFILL SHALL BE FREE OF UNSUITABLE MATERIAL SUCH AS LARGER ROCK, MUCK, AND DEBRIS.
5. DENSITY TESTS ARE REQUIRED IN 1 FOOT LIFTS ABOVE THE PIPE AT INTERVALS OF 400' MAX., OR AS DIRECTED BY THE INSPECTOR.
6. STREET PAVEMEMT 2" ASPHALT, BASE 8" LIMEROCK.

Pavement Replacement Detail



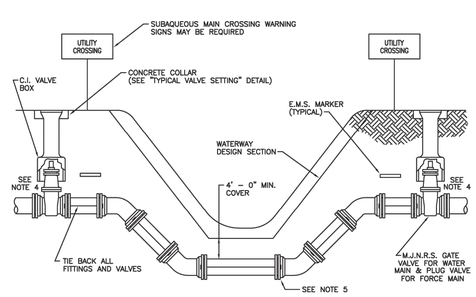
SECTION A-A



SEWER-WATER MAIN CROSSING DETAIL

STANDARD WATER AND SEWER SEPARATION STATEMENT

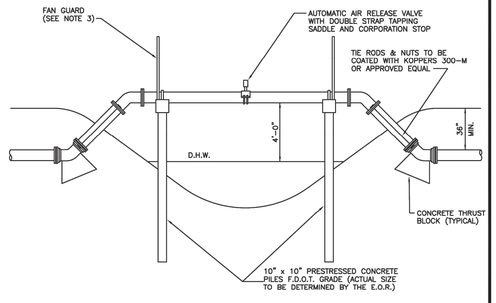
1. STORM AND SANITARY SEWER CROSSING UNDER WATER MAINS SHALL BE LAID TO PROVIDE A MINIMUM VERTICAL DISTANCE OF EIGHTEEN (18) INCHES BETWEEN THE INVERT OF THE UPPER PIPE AND THE LOWER PIPE. WHERE THIS MINIMUM SEPARATION CANNOT BE MAINTAINED, THE CROSSING SHALL BE ARRANGED SO THAT THE SEWER PIPE JOINTS AND WATER MAIN JOINTS ARE EQUIDISTANT FROM THE POINT OF CROSSING WITH NO LESS THAN TEN (10) FEET BETWEEN ANY TWO JOINTS AND BOTH PIPES SHALL BE D.I.P. WHERE THERE IS NOT ALTERNATIVE TO SEWER PIPES CROSSING OVER A WATER MAIN, THE CRITERIA FOR MINIMUM SEPARTION BETWEEN LINES AND JOINTS IN THE ABOVE, SHALL BE REQUIRED AND BOTH PIPES SHALL BE D.I.P. IRRESPECTIVE OF SEPARATION.
2. MAINTAIN TEN (10) FEET HORIZONTAL DISTANCE BETWEEN WATER MAIN AND STORM OR SANITARY SEWER MAIN, AS A MINIMUM.
3. FORCE MAIN CROSSING WATER MAIN SHALL BE LAID TO PROVIDE A MINIMUM VERTICAL DISTANCE OF EIGHTEEN (18) INCHES BETWEEN THE OUTSIDE OF THE FORCE MAIN AND OUTSIDE OF THE WATER MAIN WITH WATER MAIN CROSSING OVER THE FORCE MAIN.



NOTES:

1. IN SOME CASES AIR RELEASE VALVES MAY BE REQUIRED.
2. EXACT LOCATION AND NUMBER BLOWOFFS, VALVES, SIGNS, AND E.M.S. MARKERS TO BE DETERMINED IN THE FIELD.
3. IF WATERWAY IS NOT CUT TO THE DESIGN CROSS SECTION, THE 4'-0" MINIMUM COVER SHALL APPLY TO THE DEEPER SECTION, WHETHER EXISTING OR DESIGN.
4. APPROVED SELF RESTRAINING GASKETS SHALL BE USED ON ALL PUSH ON PIPE FOR A DISTANCE OF SIXTY (60) FEET FROM EACH VALVE.
5. APPROVED RESTRAINED MECHANICAL JOINTS SHALL BE USED FROM VALVE TO VALVE.

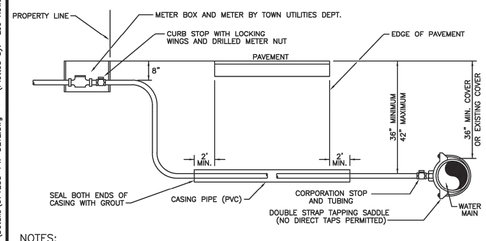
SUB-AQUEOUS CROSSING
N.T.S.



NOTES:

1. ALL EXPOSED PIPE SHALL BE DUCTILE IRON WITH FLANGED FITTINGS. RETAINER GLANDS AND UNIFLANGE TYPE FITTINGS ARE NOT TO BE SUBSTITUTED FOR FLANGED FITTINGS.
2. SPAN LENGTHS AS REQUIRED BY PERMITTING AGENCY.
3. FAN GUARDS ARE REQUIRED. SEE "TYPICAL FAN GUARD" DETAIL.
4. PIPE SHALL BE CRADLED ON NEOPRENE, 1/2" THICK MINIMUM.
5. TIE-DOWN STRAPS SHALL FIT PROPERLY AND SECURE PIPE IN CRADLE.
6. PIPE CRADLE IN CAP SHALL CONTACT 1/2 CIRCUMFERENCE OF PIPE.
7. SHOW ULTIMATE CANAL SECTION AND RELEVANT ELEVATIONS AND DISTANCES ON THIS DETAIL.
8. STAINLESS STEEL (316) REQUIRED FOR ALL STRAPS, SADDLES, FLANGE BOLTS, AND OTHER HARDWARE FOR INSTALLATIONS OVER BRACKISH OR MARINE WATERS (ANTI-GALL COMPOUND TO BE USED WHEN ASSEMBLING STAINLESS STEEL NUTS AND BOLTS.)
9. LENGTH OF SPAN WILL DETERMINE NUMBER OF PILES REQUIRED.
10. AERIAL CROSSING TO BE FIELD COATED PER F.D.O.T. SPECIFICATIONS FOR PIPING ATTACHED TO BRIDGE STRUCTURES. COLOR SHALL BE LT GREY ANSI 70 TNEMEC.

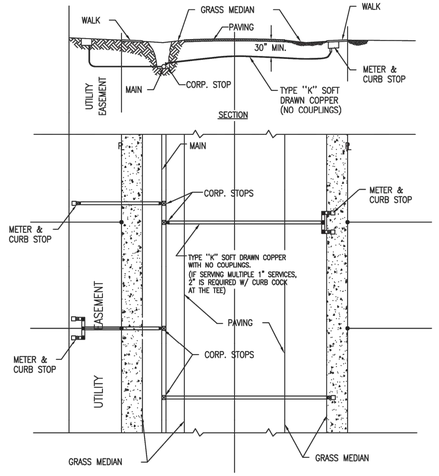
AERIAL CANAL CROSSING



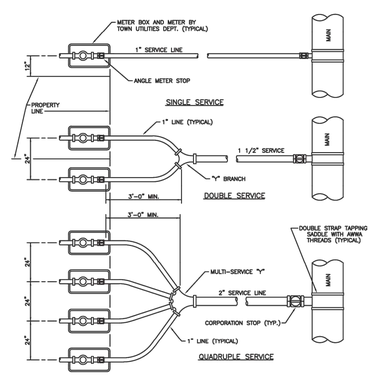
NOTES:

1. CASINGS SHALL BE REQUIRED FOR ALL LONG SIDE SERVICES.
2. SUCCESSIVE TAPS INTO THE WATER MAIN SHALL BE SPACED A MINIMUM OF 18" OFFSET.
3. WHERE NO SIDEWALK EXISTS, METER BOXES SHALL BE SET TO CONFORM TO FINISH GRADE.
4. COPPER TUBING SHALL BE TYPE "K" WITH COMPRESSION FITTINGS.
5. USE EMS MARKER #1252 COLOR BLUE DIRECTLY OVER THE CORPORATION STOP.
6. ALL SERVICE LINES SHALL BE EQUIPED WITH A CORPORATION STOP AT THE MAIN AND A CURB STOP (LOCKING CONNECTION TYPE) AT THE METER.
7. THE COPPER SERVICE LINE SHALL BE CONTINOUS FROM CORPORATION STOP TO CURB STOP WITH NO FITTINGS IN BETWEEN.
8. TAPPING SADDLES AND CORPORATION STOPS SHALL HAVE AWWA INLET THREADS.
9. ALL EXPOSED FITTINGS TO BE COATED WITH COLA TAR EPOXY.
10. GALVANIZED CASING REQUIRED FOR ANY INSTALLATION REQUIRING A JACK AND BORE, SCH. 40 PVC MAY BE USED FOR AN OPEN CUT INSTALLATION WITH THE APPROVAL OF THE TOWN UTILITIES DEPT. CASING SHOULD EXTEND TWO (2) FEET BEYOND EDGE OF PAVEMENT AND SIZED AS FOLLOWS:
    A.) 1" SERVICE USE 2" CASING
    B.) 1 1/2" SERVICE USE 3" CASING
    C.) 2" SERVICE USE 4" CASING

3/4" - 2" TYPICAL WATER SERVICE INSTALLATION



TYPICAL DOMESTIC SERVICE INSTALLATIONS

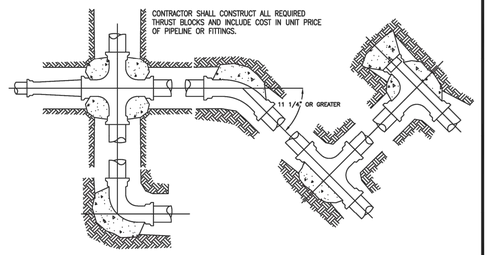


NOTES:

1. AUTHORIZED SERVICE LINE MATERIAL:
    A) TYPE "K" COPPER TUBING WITH COMPRESSION FITTINGS.
    B) POLYETHYLENE TUBING SDR 9, COPPER TUBE SIZE.
2. ANGLE METER STOP SHALL BE 1" MINIMUM.
3. 1" ANGLE METER STOPS WITH 3/4" VALVES SHALL NOT BE PERMITTED.
4. MULTIPLE SERVICE/METER INSTALLATIONS OF MORE THAN 4 METERS PER SERVICE AND SERVICE LINES LARGER THAN 2" DIAMETER SHALL BE HANDLED ON AN INDIVIDUAL BASIS.
5. ANGLE METER STOPS 1 1/2" AND 2" IN SIZE SHALL BE PROVIDED WITH BOTH A LOCKING CAP AND METER FLANGE.
6. NO FITTINGS BETWEEN CORP STOP AND ANGLE METER STOP ALLOWED WHEN USING POLYETHYLENE TUBING.

TYPICAL WATER SERVICES

TYPICAL LOCATIONS WHICH REQUIRE CONCRETE REACTION (THRUST) BLOCKS, FOR PRESSURE MAINS 4"SIZE AND GREATER CONCRETE SHALL HAVE 2500 PSI MINIMUM STRENGTH AT 28 DAYS AND BEAR AGAINST UNDISTURBED STABLE SOILS. AREA OF CONTACT SHALL BE GOVERNED BY PIPE SIZE, MAX. PRESSURE IN PIPE AND BEARING CAPACITY OF SOIL.



PIPE	THRUST BLOCK AREA REQ'D	PIPE	THRUST BLOCK AREA REQ'D	REMARKS
4"	2.0 SQ. FT.	18"	30.0 SQ. FT.	VALUES ARE FOR 90° BEND, BASED ON 2000 PSF SAFE BEARING LOAD AND PIPE PRESSURE OF 150 PSI PLUS 33% SAFETY FACTOR FOR OTHER SOILS & PRESSURES THE AREA REQUIRED IS IN DIRECT PROPORTION
6"	4.0 SQ. FT.	20"	37.0 SQ. FT.	
8"	6.6 SQ. FT.	24"	53.0 SQ. FT.	
10"	10.0 SQ. FT.	27"	80.0 SQ. FT.	
12"	14.0 SQ. FT.	30"	98.0 SQ. FT.	
14"	18.6 SQ. FT.	36"	127.0 SQ. FT.	
16"	24.0 SQ. FT.			

NOTE: FOR OTHER FITTINGS USE FOLLOWING FACTORS

TEE 100%
45° BEND 76%
22 1/2° BEND 39%
11 1/4° BEND 20%
DEAD END 100%

CONCRETE THRUST BLOCKS

NO	DATE	REVISION	BY

DESIGNED P.A.F. DATE JUNE 2011
DRAWN L.C.R. DATE JUNE 2011
CHECKED P.A.F. DATE JUNE 2011

TOWN OF MANGONIA PARK UTILITIES
1755 East Tiffany Drive, Mangonia Park, Florida 33407
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POTABLE WATER DISTRIBUTION SYSTEM
TOWN OF MANGONIA PARK

DETAILS

SCALE AS NOTED
PROJECT No. 07-1233.5
CAD FILE 071233-PW-DETL

SHEET PW-1
OF

File Name: Z:\2007\071233 Town Of Mangonia Park\071233.5 General Water And Sewer\2400\Details\071233-PW-DETL.dwg -- (Plotted by: Lee Rowbotham on Tuesday, January 14, 2020 11:03:51 AM)

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STATE OF FLORIDA

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FIELD:
DRAWN: P. Saffold
DESIGNED: TNM
APPROVED: TNM
PROJECT #21-030

NO. DATE REVISIONS

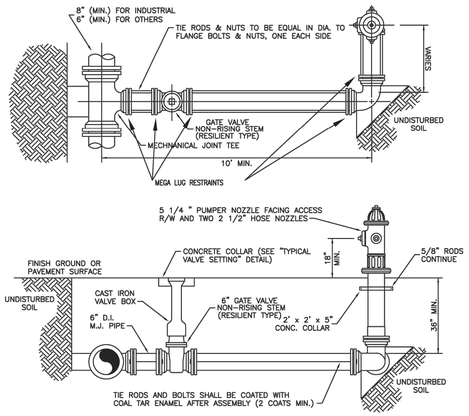
WATER & WASTEWATER DETAILS
1319 MANGONIA PARK WAREHOUSE
MANGONIA PARK, FLORIDA

TOWN RESUBMITTAL
DATE: 8/1/2024

SHEET
C4.1
OF 13

ALL ELEVATIONS REFER TO NORTH AMERICAN VERTICAL DATUM OF 1988 (NAVD 88). TO CONVERT TO NGVD 29 DATUM, ADD 1.54' TO NAVD ELEVATIONS.

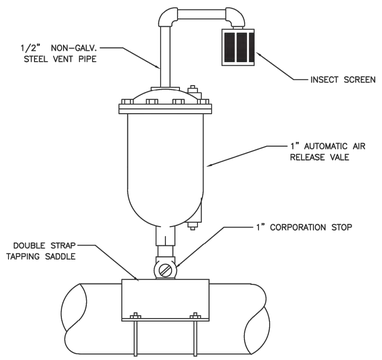
NOTE: THESE PLANS ARE PRELIMINARY AND SHOULD NOT BE USED FOR CONSTRUCTION OR BIDDING/CONTRACTUAL PURPOSES UNTIL ALL PERMITS ARE ISSUED.



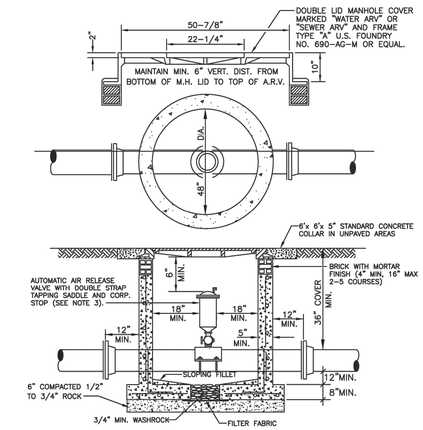
NOTES:

1. HYDRANTS SHALL BE INSTALLED PLUMB AND TRUE.
2. HYDRANTS SHALL BE PAINTED PER TOWN OF MANGONIA PARK STANDARDS:
   RED BODY REFLECTIVE AND WHITE CAP REFLECTIVE
3. HEEL AND THRUST BLOCKS TO REST IN UNDISTURBED SOIL.
4. THE ONLY FIRE HYDRANTS ACCEPTABLE ARE:
   A) KENNEDY
   B) AMERICAN–DARLING
   C) MUELLER BREAK–A–WAY
5. MUST USE MEGA LUG RESTRAINTS

FIRE HYDRANT INSTALLATION DETAIL



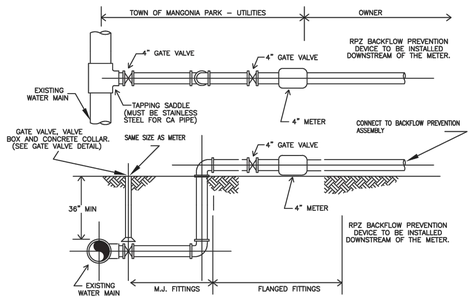
AERIAL AIR RELEASE VALVE



NOTES:

1. 4000 P.S.I. TYPE II CONCRETE
2. VAULT SHALL BE PRECAST, POURED IN PLACE CONCRETE OR CONCRETE BLOCK, ALL WITH STEEL REINFORCING.
3. AIR RELEASE VALVE SHALL BE TYPE AND SIZE APPROPRIATE FOR SERVICE INTENDED. FORCE MAINS SHALL BE 2" MIN.
4. ALL OPENINGS SHALL BE SEALED WITH WATERPROOF NON–SHRINKING GROUT.
5. MANHOLE FABRICATION SHALL BE IN ACCORDANCE WITH ASTM C–478 LATEST STANDARDS.
6. PAINT THE INSIDE AND OUTSIDE WITH TWO (2) COATS OF AN APPROVED PROTECTIVE COATING.
7. CONCRETE COLLAR REQUIRED WHEN MANHOLE IS OUTSIDE PAVEMENT.

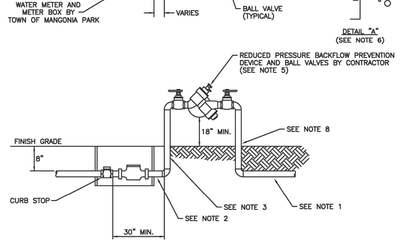
AIR RELEASE VALVE & VAULT



NOTES:

1. ALL SERVICE PIPING SHALL BE DI PIPE THE SAME SIZE AS METER.
2. ALL FITTINGS FLANGED WITH MEGALUGS OR EQUAL.
3. METER SERVICE SHALL NOT BE INSTALLED UNTIL:
   A – FORM BOARDS FOR DRIVEWAY AND/OR SIDEWALK ARE IN PLACE, OR
   – DIRVEWAY OR SIDEWALK IS IN PLACE, OR
   – "FINISHED GRADE" STAKE IS SET ADJACENT TO PROPOSED METER BOX LOCATION.
   B – THE REQUIRED BACKFLOW PREVENTION DEVICE IS INSTALLED.
4. INSTALLATION OF THE DOMESTIC SERVICE METER AND METER BOX WILL BE PERFORMED BY THE TOWN OF MANGONIA PARK–UTILITIES.
   THE DOMESTIC TAP ON TH EXISTING MAIN HAS TO BE DONE BY THE CONTRACTOR IF THE TAP IS 4 " AND ABOVE .THE OWNER SHALL BE RESPONSIBLE FOR ALL INSTALLATION AND MAINTAINANCE DOWNSTREAM OF THE DOMESTIC METER.

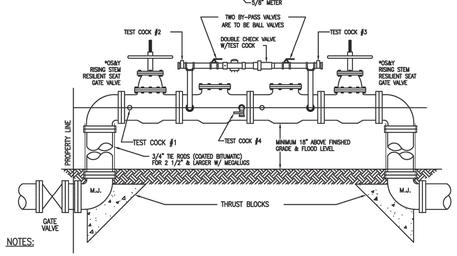
3",4",6",8",10" WATER METER DETAIL



NOTES:

1. TYPE "K" COPPER TUBING (SOFT DRAWN) OR POLYETHYLENE TUBING SDR 9 COPPER TUBE SIZE.
2. IF THIS DISTANCE IS LESS THAN 3 FEET ONLY COPPER TUBING SHALL BE PERMITTED.
3. 3/4"–2" BOTH RISERS SHALL BE TYPE "K" COPPER TUBING (HARD DRAWN) WITH COPPER/BRASS SOLDER FITTINGS AND ADAPTERS. ONLY LEAD FREE SOLDER AND FLUX SHALL BE PERMITTED.
4. BOTH 1 1/2" AND 2" ANGLE METER STOPS SHALL BE EQUIPPED WITH LOCKING CAPS AND METER FLANGES.
5. ALL BACKFLOW PREVENTION DEVICES ASSEMBLY MUST BE USC APPROVED.
6. WHEN THE DEVICE IS INSTALLED PARALLEL TO A BUILDING WALL THERE SHALL BE A MIN. OF 6" BETWEEN THE EDGE OF THE DEVICE AND THE BUILDING WALL.
7. 2 1/2 "–10" THE RPZ BACKFLOW PREVENTION ASSEMBLY MUST BE INSTALLED DOWNSTREAM OF THE WATER SERVICE METER 18"–24" ABOVE FINISHED GRADE ON NON–GALVANIZED METAL PIPE. WORK DONE ON THIS DEVICE MUST BE PERFORMED BY A CERTIFIED BACKFLOW TECHNICIAN AND TEST REPORTS MUST BE SUBMITTED TO THE TOWN OF MANGONIA PARK UTILITIES.

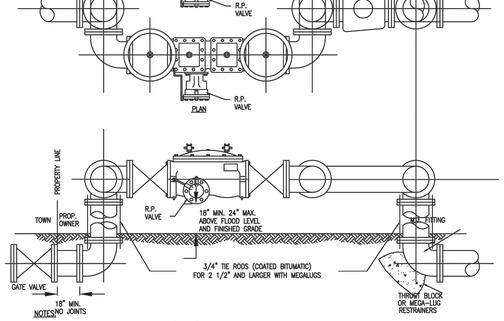
RPZ BACKFLOW PREVENTION DEVICE



NOTES:

1. ABOVE GROUND INSTALLATION OF 3" & ABOVE REQUIRE RISING STEM O.S. & Y. RESILIENT SEAT GATE VALVES WITH TEST COCKS.
2. INSTALLATION AND CERTIFICATION REQUIRED BY A CERTIFIED BACKFLOW TECHNICIAN PRIOR TO ACCEPTANCE.
3. AN RPZ BACKFLOW PREVENTION ASSEMBLY MUST BE INSTALLED DOWNSTREAM OF THE DOMESTIC AND IRRIGATION METER, 18" – 24" ABOVE FINISHED GRADE ON NON–GALVANIZED METALLIC PIPE. WORK DONE ON THIS DEVICE IS TO BE PERFORMED BY A CERTIFIED BACKFLOW TECHNICIAN AND TEST REPORTS MUST BE SUBMITTED TO THE TOWN OF MANGONIA PARK UTILITIES.

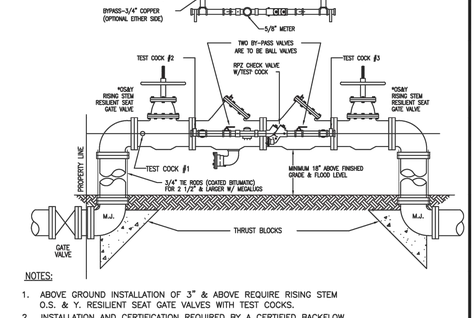
DOUBLE DETECTOR CHECK ASSEMBLY & METER INSTALLATION SIZES:3",4",6",8",10"



NOTES:
NO JOINTS

1. ABOVE GROUND INSTALLATION OF 3/4" TO 2" REQUIRE BALL VALVES WITH TEST COCKS.
2. ALL PIPING PRIOR TO BACKFLOW PREVENTION ASSEMBLIES SHALL BE EITHER CEMENT LINED DUCTILE IRON PIPE OR COPPER FOR THE SMALLER DEVICES. THIS INCLUDES DOMESTIC, IRRIGATION AND FIRE LINE DEVICES. ALL PIPING SHALL MEET THE TOWN OF MANGONIA PARK PLUMBING AND MECHANICAL CODES.
3. AN RPZ BACKFLOW PREVENTION ASSEMBLY MUST BE INSTALLED DOWNSTREAM OF THE DOMESTIC AND IRRIGATION METER, 18" – 24" ABOVE FINISHED GRADE ON NON–GALVANIZED METALLIC PIPE. WORK DONE ON THIS DEVICE MUST BE PERFORMED BY A CERTIFIED BACKFLOW TECHNICIAN AND TEST REPORTS MUST BE SUBMITTED TO THE TOWN OF MANGONIA PARK UTILITIES

PARALLEL REDUCED PRESSURE BACKFLOW PREVENTION ASSEMBLIES
N.T.S.



NOTES:

1. ABOVE GROUND INSTALLATION OF 3" & ABOVE REQUIRE RISING STEM O.S. & Y. RESILIENT SEAT GATE VALVES WITH TEST COCKS.
2. INSTALLATION AND CERTIFICATION REQUIRED BY A CERTIFIED BACKFLOW TECHNICIAN PRIOR TO ACCEPTANCE.
3. AN RPZ BACKFLOW PREVENTION ASSEMBLY MUST BE INSTALLED DOWNSTREAM OF THE DOMESTIC AND IRRIGATION METER, 18" – 24" ABOVE FINISHED GRADE ON NON–GALVANIZED METALLIC PIPE. WORK DONE ON THIS DEVICE IS TO BE PERFORMED BY A CERTIFIED BACKFLOW TECHNICIAN.
4. TEST REPORTS MUST BE SUBMITTED TO THE TOWN OF MANGONIA PARK – UTILITIES AFTER INSTALLATION

REDUCED PRESSURE DETECTOR ASSEMBLY
SIZES – 3",4",6",8", & 10"

NO	DATE	REVISION	BY	NO	DATE	REVISION	BY

DESIGNED P.A.F. DATE JUNE 2011
DRAWN L.C.R. DATE JUNE 2011
CHECKED P.A.F. DATE JUNE 2011

TOWN OF MANGONIA PARK UTILITIES
1755 East Tiffany Drive, Mangonia Park, Florida 33407
Phone: 561.848.1235 Fax: 561.848.6940

POTABLE WATER DISTRIBUTION SYSTEM
TOWN OF MANGONIA PARK

DETAILS

SCALE AS NOTED
PROJECT No 07–1233.5
CAD FILE 071233–PW–DETL

SHEET PW-2
OF

File Name: Z:\2007\071233 Town Of Mangonia Park\071233.5 General Water And Sewer\CADD\Details\071233–PW–DETL.dwg – (Plotted by: Lee Rowbotham on Tuesday, January 14, 2020 11:12:03 AM)

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FIELD:
DRAWN: P. Saffold
DESIGNED: TNM
APPROVED: TNM
PROJECT #21-030

NO. DATE REVISIONS

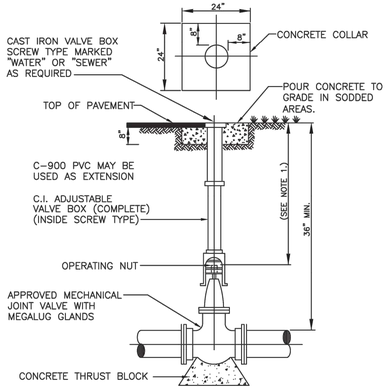
WATER & WASTEWATER DETAILS
1319 MANGONIA PARK WAREHOUSE
[redacted]
MANGONIA PARK, FLORIDA

TOWN RESUBMITTAL
DATE: 8/1/2024

SHEET
C4.2
OF 13

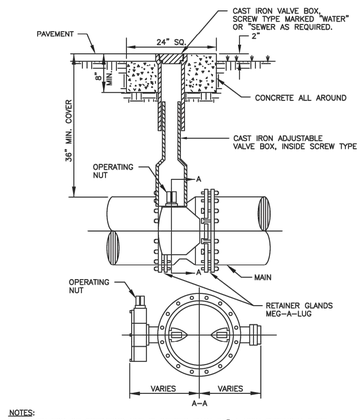
ALL ELEVATIONS REFER TO NORTH AMERICAN VERTICAL DATUM OF 1988 (NAVD 88). TO CONVERT TO NGVD 29 DATUM, ADD 1.54' TO NAVD ELEVATIONS.

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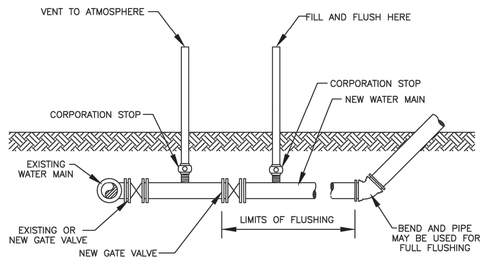
NOTES:
1. WHEN OPERATING NUT IS DEEPER THAN 36" AN EXTENSION WILL BE REQUIRED TO BRING OPERATING NUT 24"–30" BELOW FINISHED GRADE. EXTENSION BOLTS & NUTS ARE TO BE STAINLESS STEEL. A STAINLESS STEEL CENTERING PLATE, WELDED TO THE EXTENSION, IS ALSO REQUIRED.
2. VALVE BOXES SHALL HAVE LOCKING COVERS IN HIGH TRAFFIC ROADWAY AREAS.

TYPICAL GATE VALVE SETTINGS
N.T.S.



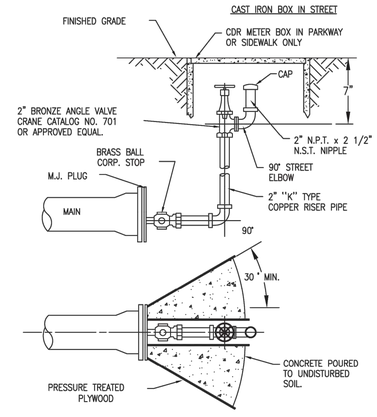
NOTES:
1. WHEN TOP OF OPERATING NUT IS DEEPER THAN 36" A HIGH STRENGTH STEEL EXTENSION WILL BE REQUIRED TO BRING OPERATING NUT 24"–30" BELOW FINISHED GRADE. EXTENSION BOLTS & NUTS ARE TO BE STAINLESS STEEL. A STEEL CENTERING PLATE WELDED TO THE EXTENSION, IS ALSO REQUIRED.
2. VALVE BOXES IN PAVEMENT SHALL HAVE LOCKING COVERS MARKED "WATER" OR "SEWER" AS REQUIRED.

TYPICAL BUTTERFLY VALVE SETTING



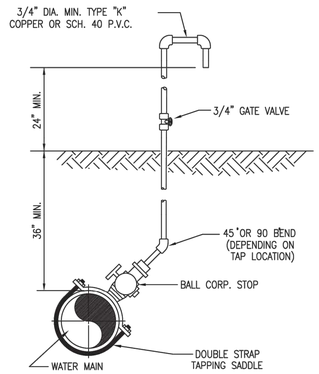
NOTES:
1. DO NOT REMOVE TEMPORARY CONNECTION AT CORPORATION STOPS ON NEW MAIN UNTIL FILLING AND FLUSHING AND TESTING HAS BEEN COMPLETED.
2. PLUG CORPORATION STOP WITH A P.V.C. PLUG WHEN ABANDONED.
3. PROVIDE ALL NECESSARY THRUST BLOCKS AND/OR OTHER RESTRAINTS.
4. THIS IS THE ONLY FLUSHING METHOD APPROVED BY THE TOWN OF MANGONIA PARK UTILITIES. SINCE OTHER METHODS DO NOT REMOVE LARGE OBJECTS SUCH AS ROCKS, CANS, BOTTLES AND OTHER DEBRIS FROM MAINS. ALL FIRE HYDRANTS AND DETECTOR CHECK VALVES MUST BE INSPECTED AND TESTED TO ASSURE NO SUCH FOREIGN OBJECTS ARE IN THE FIRE FLOW.
5. THE GENERAL CONTRACTOR MUST RENT A FIRE HYDRANT METER FROM THE TOWN OF MANGONIA PARK UTILITIES FOR USE OF POTABLE WATER FROM A TOWN OWNED SOURCE.

FILLING & FLUSHING DETAIL
N.T.S



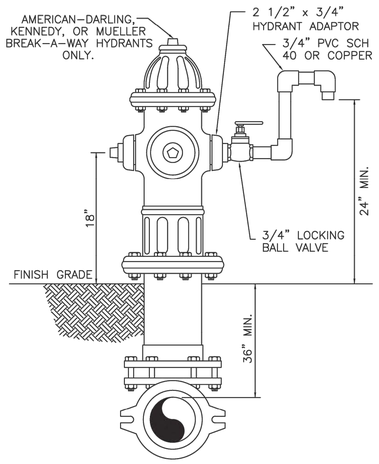
NOTES:
1. PIPE JOINT COMPOUND SHALL BE APPLIED TO MALE THREADS ONLY.
2. COAT ALL EXPOSED THREADS WITH BITUMASTIC BEFORE BACKFILLING.
3. PLYWOOD AND CONCRETE TO HAVE A HEIGHT EQUAL TO THE DIAMETER OF THE PIPE.
4. CONCRETE STRENGTH – 4 KSI @ 28 DAYS.
5. GALVINIZED PIPING AND FITTINGS SHALL NOT BE USED ON TEMPORARY OR PERMANENT BLOWOFFS.

2" TERMINAL BLOWOFF
N.T.S.

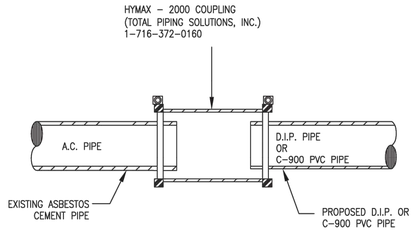


NOTE:
WHENEVER POSSIBLE, SERVICE TAPS SHALL BE USED AS SOMPLE POINTS.

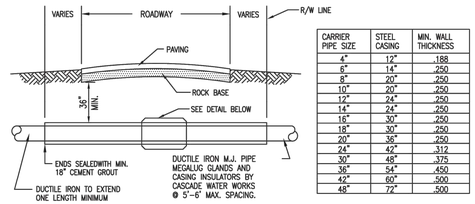
SAMPLE POINT (ON MAIN)
N.T.S.



SAMPLE POINT (HYDRANT)
N.T.S.



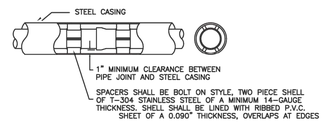
PIPE TRANSITION DETAIL



CARRIER PIPE SIZE	STEEL CASING	MIN. WALL THICKNESS
4"	12"	.188
6"	14"	.250
8"	20"	.250
10"	20"	.250
12"	24"	.250
14"	24"	.250
16"	30"	.250
18"	30"	.250
20"	36"	.250
24"	42"	.312
30"	48"	.375
36"	54"	.450
42"	60"	.500
48"	72"	.500

NOTES:
1. APPROVED SELF RESTRAINING GASKETS SHALL BE USED FOR 4" THRU 12" DIAMETER PIPE.
2. RESTRAINED MECHANICAL JOINTS SHALL BE USED FOR PIPE 14" DIAMETER AND ABOVE AND SHALL HAVE FACTORY WELDED RETAINING RINGS. AMERICAN RESTRAINED JOINT PIPE OR APPROVED EQUAL MAY BE USED.
3. INSTALL STAINLESS STEEL PIPE CASING SPACERS PER MANUFACTURERS RECOMMENDATIONS (CASCADE MFG. CO. OR APPROVEDEQUAL).
4. PIPE CASING SPACERS SHALL BE POSITIONED PER MANUFACTURERS RECOMMENDATIONS.

STANDARD JACK & BORE DETAIL
N.T.S.



CASING DETAIL
N.T.S.

ALL ELEVATIONS REFER TO NORTH AMERICAN VERTICAL DATUM OF 1988 (NAVD 88). TO CONVERT TO NGVD 29 DATUM, ADD 1.54' TO NAVD ELEVATIONS.

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NO	DATE	REVISION	BY

DESIGNED P.A.F. DATE JUNE 2011
DRAWN L.C.R. DATE JUNE 2011
CHECKED P.A.F. DATE JUNE 2011

TOWN OF MANGONIA PARK UTILITIES
1755 East Tiffany Drive, Mangonia Park, Florida 33407
Phone: 561.848.1235 Fax: 561.848.6940

POTABLE WATER DISTRIBUTION SYSTEM
TOWN OF MANGONIA PARK

DETAILS

SCALE AS NOTED
PROJECT No 07-1233.5
CAD FILE 071233-PW-DETL

SHEET PW-3
OF

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SEAL: TODD N. MCLEOD, P.E. LICENSE NO. 81128 STATE OF FLORIDA

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FIELD:
DRAWN: P. Saffold
DESIGNED: TNM
APPROVED: TNM

PROJECT #21-030

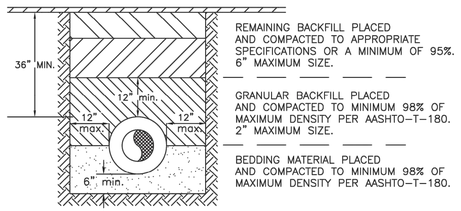
NO. DATE REVISIONS

WATER & WASTEWATER DETAILS
1319 MANGONIA PARK WAREHOUSE
MANGONIA PARK, FLORIDA

TOWN RESUBMITTAL
DATE: 8/1/2024

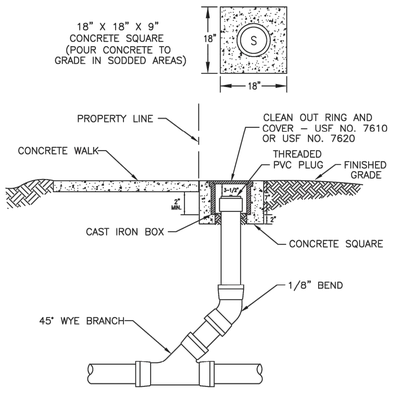
SHEET C4.3
OF 13

File Name: Z:\2007\071233 Town Of Mangonia Park\071233.5 General\ Water And Sewer\CADD\Details\071233-PW-DETL.dwg — (Plotted by: Lee Rowbothom on Tuesday, January 14, 2020 11:20:23 AM)



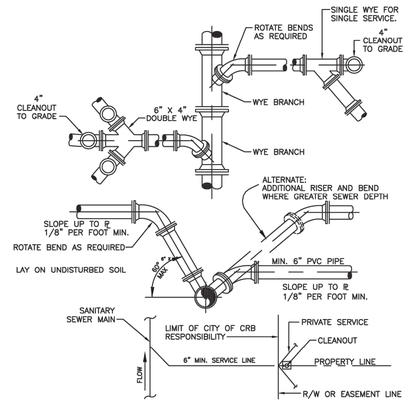
NOTES:
1. BEDDING SHALL CONSIST OF IN-SITU GRANULAR MATERIAL OR WASHED AND GRADED LIMEROCK 3/8"-7/5" SIZING.
2. THE PIPING SHALL BE FULLY SUPPORTED FOR ITS ENTIRE LENGTH WITH APPROPRIATE COMPACTION UNDER THE PIPE HAUNCHES.
3. THE PIPE SHALL BE PLACED IN A DRY TRENCH.
4. BACKFILL SHALL BE FREE OF UNSUITABLE MATERIAL SUCH AS LARGER ROCK, MUCK, AND DEBRIS.
5. DENSITY TESTS ARE REQUIRED IN 1 FOOT LIFTS ABOVE THE PIPE AT INTERVALS OF 400' MAX., OR AS DIRECTED BY THE INSPECTOR.
6. STREET PAVEMEMT 2" ASPHALT, BASE 8" LIMEROCK.

PAVEMENT REPLACEMENT DETAIL
N.T.S.



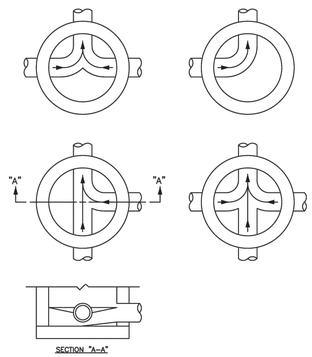
NOTES:
1. CLEAN OUTS TO BE LOCATED IN GRASS WHENEVER POSSIBLE.
2. CLEANOUTS SHALL NOT BE INSTALLED IN TRAFFIC LANES OR AREAS UNDER HEAVY TRAFFIC LOADS.
3. THE COVER TO BE MARKED "S".
4. CLEANOUTS TO BE INSTALLED PRIOR TO WATER METER RELEASE.
5. NO 90 BENDS SHALL BE USED FOR CLEANOUT INSTALLATION.

CLEAN OUT AT PROPERTY LINE
N.T.S.



NOTES:
1. SERVICE LATERALS SHALL TERMINATE AT PROPERTY LINE AT A DEPTH OF 3 FEET.
2. CLEANOUT CONNECTION TO BE EXTENDED 3 FEET ABOVE GRADE PRIOR TO WATER METER INSTALLATION.
3. CLEANOUTS ARE TO BE SET TO GRADE PRIOR TO WATER METER ACTIVATION.
4. NO 90° BENDS SHALL BE PERMITTED FOR SEWER SERVICE AND CLEANOUT INSTALLATION.

TYPICAL LATERAL CONNECTIONS
N.T.S.



NOTE:
1. ALL INVERT CHANNELS ARE TO BE CONSTRUCTED FOR SMOOTH FLOW WITHOUT OBSTRUCTION.
2. PROPERLY SHAPED SPILLWAYS SHALL BE CONSTRUCTED BETWEEN PIPES WITH DIFFERENT INVERT ELEVATIONS TO PROVIDE FOR SMOOTH FLOWS.
3. SERVICE LATERALS ENTERING MANHOLES SPECIFIED ON PLANS MUST HAVE FLOW CHANNELS.
4. BRICKS AND CONCRETE RUBBLE PERMITTED FOR FLOW CHANNEL BUILDUP.
5. SIDEWALLS OF FLOW CHANNEL SHALL BE AT LEAST HALF OF PIPE HEIGHT AT ALL POINTS.

TYPICAL INVERT FLOW CHANNELS
N.T.S.

NOTE:
1. FRAME AND COVER SHALL BE U.S. FOUNDRY NO. 402A-AK, VULCAN NO. 1344 OR APPROVED EQUAL FOR 1" DEPTH FRAME, OR U.S. FOUNDRY NO. 465A OR EQUAL FOR 4" DEPTH FRAME.
2. A STEEL MANHOLE RISER (MANHOLE SYSTEMS MODEL MS101) SHALL BE USED TO ELEVATE EXISTING MANHOLE COVERS TO RESURFACE GRADE (MAX. 4" HEIGHT).
3. MANHOLE CASTINGS SHALL BE WATER TIGHT.
4. ALL BEARING SURFACES TO BE MACHINED (IRON TO IRON).
5. FRAME WT.: MIN. 250 LBS. COVER WT.: MIN. 160 LBS.
6. SPECIFY: SANITARY FOR SEWER OR WATER FOR POTABLE WATER

TYPICAL MANHOLE COVER
N.T.S.

NOTES:
1. PRECAST CONCRETE, TYPE II, ACID RESISTANT, 4000 P.S.I.
2. RAMNEK FLEXIBLE PLASTIC GASKET OR EQUAL AT ALL RISER JOINTS (1/2" THICK, WIDTH AT LEAST HALF THE WALL THICKNESS) WITH NON-SHRINK WATERPROOF GROUT ON INSIDE AND OUTSIDE.
3. ALL OPENINGS SHALL BE SEALED WITH EITHER RESILIENT CONNECTOR (BOOTS) OR P.V.C. MANHOLE ADAPTERS FOR P.V.C. PIPE PENETRATIONS. BOOTS AND ADAPTERS SHALL BE CEMENTED IN WITH NON-SHRINK WATERPROOF GROUT.
4. FLOW CHANNELS SHALL BE CONSTRUCTED TO DIRECT INFLUENT INTO FLOW STREAM, AND SHALL BE FORMED IN ALL MANHOLES TO ACCEPT A T.V. CAMERA.
5. LIFT HOLES ARE PERMITTED, BUT SHALL BE SEALED.
6. ALL PIPE HOLES SHALL BE PRECAST OR CORE-DRILLED, WITH NON-SHRINK, WATER PROOF GROUT.
7. INSIDE DROPS SHALL NOT BE DESIGNED TO EXCEED 1.80 FT. AND NOT CONSTRUCTED TO EXCEED 2.0 FT.
8. MIN. HOLE ø EQUALS PIPE O.D. PLUS 8" FOR P.V.C. ADAPTERS OR PIPE O.D. PLUS 4" FOR BOOTS.
9. MANHOLE FABRICATION SHALL BE IN ACCORDANCE WITH ASTM C-478, LATEST STANDARD.
10. CONCRETE TO BE SEAL COATED WITH 2 COATS OF COAL EPOXY AND 2 COATS OUTSIDE.
11. MANHOLE MUST BE INSPECTED BEFORE PAINTING AND INSTALLATION.

TYPICAL MANHOLE DETAILS
N.T.S.

NOTE:
1. DROP M.H. MUST BE USED WHEN DIFFERENCE OF INVERTS IS 1'-6" USING A TEE AT THE TOP AND A 90° BEND AT THE BOTTOM AS SHOWN.
2. DETAILS NOT SHOWN ARE TO BE THE SAME AS FOR THE "TYPICAL MANHOLE DETAIL".
3. STANDARD PRECAST MANHOLE SECTION (4' ø) AND SLAB TO BE CAST MONOLITHICALLY AND SHALL FULLY EXTEND UNDER THE DROP.
4. INSIDE DROPS ALLOWED ONLY ON EXISTING MANHOLES.

TYPICAL DROP MANHOLE
N.T.S.

NOTES:
1. MANUFACTURED IN ACCORDANCE WITH ASTMC-478 LATEST REVISION.
2. 4,000 PSI CONCRETE @ 28 DAYS.

DOGHOUSE MANHOLE DETAILS
N.T.S.

NO	DATE	REVISION	BY	NO	DATE	REVISION	BY

DESIGNED P.A.F. DATE JUNE 2011
DRAWN L.C.R. DATE JUNE 2011
CHECKED P.A.F. DATE JUNE 2011

TOWN OF MANGONIA PARK UTILITIES
1755 East Tiffany Drive, Mangonia Park, Florida 33407
Phone: 561.848.1235 Fax: 561.848.6940

SANITARY SEWER COLLECTION SYSTEM TOWN OF MANGONIA PARK

DETAILS

SCALE: AS NOTED
PROJECT No: 07-1233.5
CAD FILE: 071233-SS-DETL

SHEET SS-1 OF

ALL ELEVATIONS REFER TO NORTH AMERICAN VERTICAL DATUM OF 1988 (NAVD 88). TO CONVERT TO NGVD 29 DATUM, ADD 1.54' TO NAVD ELEVATIONS.

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DRAWN: P. Saffold
DESIGNED: TNM
APPROVED: TNM
PROJECT #21-030

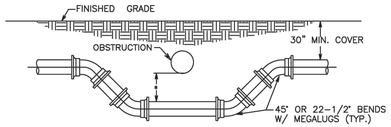
NO. | DATE | REVISIONS

WATER & WASTEWATER DETAILS
1319 MANGONIA PARK WAREHOUSE
MANGONIA PARK, FLORIDA

TOWN RESUBMITTAL
DATE: 8/1/2024

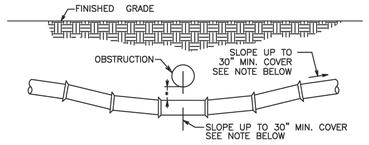
SHEET
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OF 13

File Name: Z:\2007\071233 Town Of Mangonia Park\071233.9 General Water And Sewer\CADD\Details\071233-SS-DETL.dwg — (Plotted by: Lee Rowbotham on Tuesday, January 14, 2020 11:22:36 AM)



SPECIAL UTILITY CROSSING – FITTING TYPE

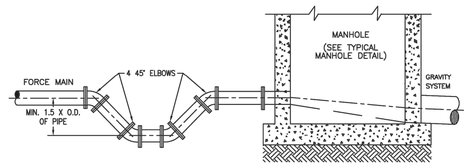
* UNLESS SHOWN OTHERWISE, 18" MIN. CLEARANCE WILL BE REQUIRED FOR WATER & SEWER MAIN CROSSING. 4" MIN. CLEARANCE WILL BE REQUIRED FOR OTHER TYPE OF UTILITIES CROSSING.



SPECIAL UTILITY CROSSING – DEFLECTION TYPE

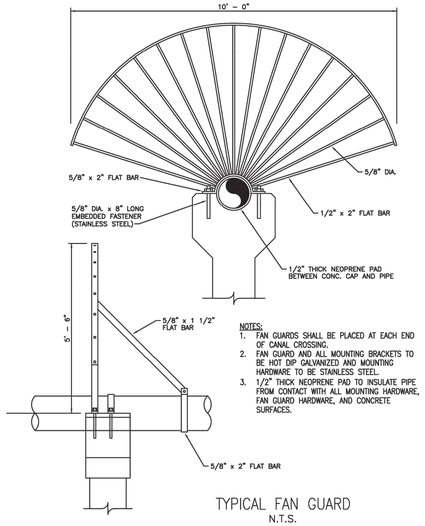
NOTES:
1. THE DEFLECTION TYPE CROSSING SHALL BE USED WHERE EVER POSSIBLE.
2. ONLY UNDER SPECIFIC ORDERS BY THE ENGINEER SHALL THE FITTING TYPE CROSSING BE ALLOWED.
3. CONSTRUCT STANDARD CROSSING USING 75% OF MANUFACTURERS MAXIMUM JOINT DEFLECTION (MAX.).
4. ALL MECHANICAL JOINTS MEGALUG.
5. FITTINGS TO BE RESTRAINED WITH RETURN GLANDS AND EITHER THRUST BLOCKS OR TIE RODS.
6. ALL EXPOSED TIE RODS SHALL BE COATED WITH COAL TAR EPOXY.

FORCE MAIN CROSSING
N.T.S.



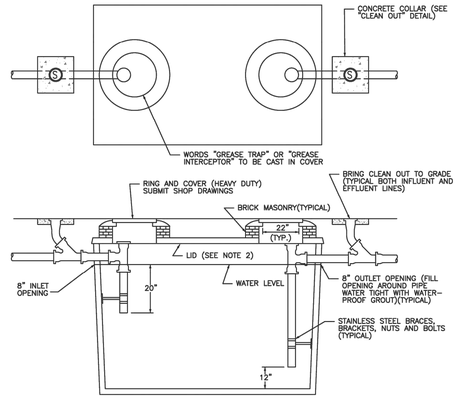
NOTES:
1. FORCE MAIN TO ENTER MANHOLE AS CLOSE AS POSSIBLE TO 180° TO GRAVITY OUTLET.
2. THE INVERT LEVEL OF FORCE MAIN AT POINT OF ENTRY SHALL BE NO MORE THAN 6" ABOVE INVERT OF MANHOLE.
3. CORE ENTRY ONLY INTO EXISTING MANHOLES.
4. TRAP TO BE LOCATED PRIOR TO DROP INTO MANHOLE AND OUTSIDE OF PAVED AREAS, IF POSSIBLE.
5. USE TWO 45° ELBOWS PAST TRAP IF ELEVATION DROP IS REQUIRED TO ENTER MANHOLE.
6. FLOW CHANNEL REQUIRED.
7. MANHOLE AND FLOW CHANNEL TO HAVE TWO DIFFERENT COLORED COATS OF 100% SOLIDS EPOXY LINER.
8. FORCE MAINS SHALL BE A MINIMUM OF 4" I.D. EPOXY-LINED D.I.P.
9. ALL MECHANICAL JOINTS TO USE MEGALUG OR EQUAL.

FORCE MAIN ENTERING MANHOLE
N.T.S.



NOTES:
1. FAN GUARDS SHALL BE PLACED AT EACH END OF CANAL CROSSING.
2. FAN GUARD AND ALL MOUNTING BRACKETS TO BE HOT DIP GALVANIZED AND MOUNTING HARDWARE TO BE STAINLESS STEEL.
3. 1/2" THICK NEOPRENE PAD TO INSULATE PIPE FROM CONTACT WITH ALL MOUNTING HARDWARE, FAN GUARD HARDWARE, AND CONCRETE SURFACES.

TYPICAL FAN GUARD
N.T.S.



NOTES:
1. SHOP DRAWINGS REQUIRED TO BE SUBMITTED FOR TOWN OF MANGONIA PARK APPROVAL.
2. LID TYPES:
   A) 4" REGULAR LID
   B) 8" TRAFFIC BEARING LID

GREASE TRAP DETAIL

PRELIMINARY – NOT FOR CONSTRUCTION

NO	DATE	REVISION	BY	NO	DATE	REVISION	BY

DESIGNED P.A.F. DATE JUNE 2011
DRAWN L.C.R. DATE JUNE 2011
CHECKED P.A.F. DATE JUNE 2011

TOWN OF MANGONIA PARK
UTILITIES
1755 East Tiffany Drive, Mangonia Park, Florida 33407
Phone: 561.848.1235 Fax: 561.848.6940

SANITARY SEWER
COLLECTION SYSTEM
TOWN OF MANGONIA PARK

DETAILS

SCALE AS NOTED
PROJECT No 07-1233.5
CAD FILE 071233.5-SS-DETL

SHEET SS-2 OF

File Name: Z:\2007\071233 Town Of Mangonia Park\071233.5 General Water And Sewer\CADD\Details\071233-SS-DETL.dwg – (Plotted by: Lee Rowbotham on Tuesday, January 14, 2020 11:24:38 AM)

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FIELD:
DRAWN: P. Saffold
DESIGNED: TNM
APPROVED: TNM

PROJECT #21-030

REVISIONS

NO.	DATE	REVISION

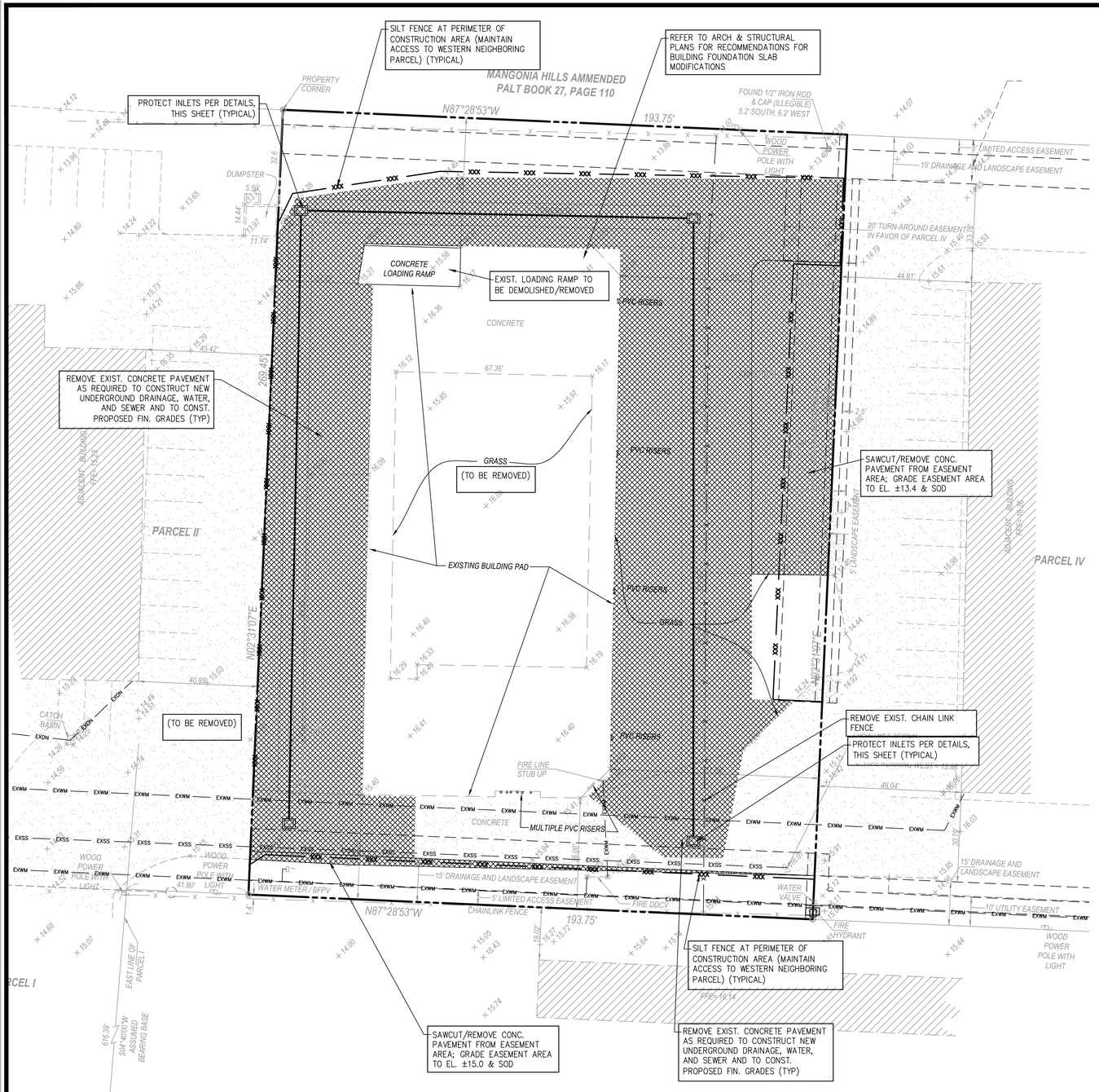
WATER & WASTEWATER DETAILS
1319 MANGONIA PARK WAREHOUSE
[redacted]
MANGONIA PARK, FLORIDA

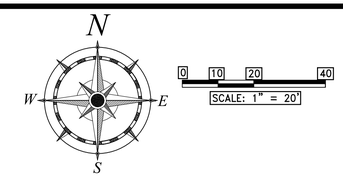
TOWN RESUBMITTAL
DATE: 8/1/2024

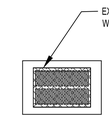
SHEET
C4.5
OF 13

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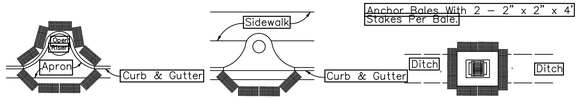
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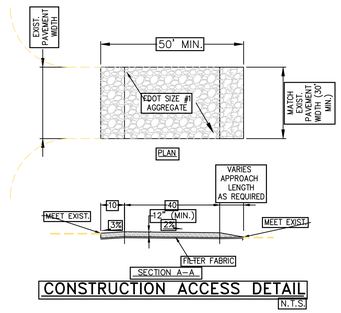




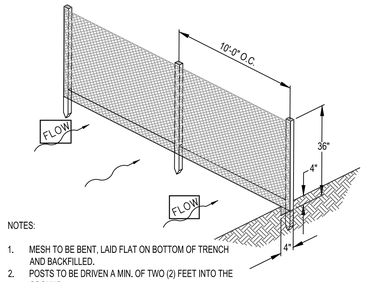
INLET PROTECTION
N.T.S.



PROTECTION AROUND INLETS OR SIMILAR STRUCTURES
CONTRACTOR TO PROTECT ALL STM STRUCTURES FROM SEDIMENTATION DURING CONSTRUCTION. N.T.S.



CONSTRUCTION ACCESS DETAIL
N.T.S.



NOTES:

1. MESH TO BE BENT, LAID FLAT ON BOTTOM OF TRENCH AND BACKFILLED.
2. POSTS TO BE DRIVEN A MIN. OF TWO (2) FEET INTO THE GROUND.
3. SILT FENCE TO BE PLACED ALL AROUND CLEARING PERIMETERS.

FILTER FABRIC SILT FENCE DETAIL
N.T.S.

NOTES:

1. PROTECT EXISTING AND PROPOSED INLETS DURING CONSTRUCTION.
2. SWEEP ROADS AND CONSTRUCTION SITE PERIODICALLY TO REMOVE CONSTRUCTION DEBRIS AND SEDIMENT.
3. STORM DRAINAGE (EXIST. & PROP.) TO BE SEDIMENT AND SILT FREE DURING CONSTRUCTION AND AT FINAL SITE WALK THROUGH.
4. ALL EXISTING CURBING, PAVEMENT, CONCRETE & BASEROCK TO BE REMOVED IN THEIR ENTIRETY WHERE INDICATED. BACKFILL WITH CLEAN, SUITABLE MATERIAL APPROPRIATE TO THE FUTURE APPLICATION AND IN COMPLIANCE WITH THE RESPECTIVE REQUIREMENTS (i.e. STRUCTURAL BUILDING FILL, LANDSCAPE FILL, PARKING LOT SUBGRADE).
5. EXISTING BASEROCK MAY BE REUSED FOR PROPOSED PARKING LOT UPON CONTRACTOR DEMONSTRATING THAT THE MATERIAL MEETS THE MINIMUM PAVEMENT SPECIFICATIONS INCLUDED IN THESE PLANS.
6. ALL EXISTING DRAINAGE PIPE & STRUCTURES WITHIN BOUNDARY TO BE REMOVED (UNLESS OTHERWISE NOTED).
7. DEMOLITION LIMITS ARE SHOWN FOR THE CONTRACTOR'S REFERENCE ONLY. ADDITIONAL DEMOLITION MAY BE REQUIRED & SHALL BE CONSIDERED INCIDENTAL TO THE CONSTRUCTION OF REQUIRED IMPROVEMENTS. IT IS THE CONTRACTOR'S RESPONSIBILITY TO FAMILIARIZE HIMSELF WITH THE SITE & ALL REQUIRED DEMOLITION PRIOR TO BID SUBMITTAL. ALL DEMOLISHED/REMOVED MATERIALS SHALL BE PROPERLY DISPOSED OF BY THE CONTRACTOR IN ACCORDANCE WITH ALL APPLICABLE REGULATORY REQUIREMENTS.
8. UTILITIES AND STORM DRAINAGE MAY NOT BE ABANDONED IN PLACE UNLESS SPECIFICALLY STATED WITHIN THESE PLANS AND PREVIOUSLY APPROVED BY THE APPLICABLE GOVERNING AGENCIES AND THE ENGINEER OF RECORD.
9. DISTURBED GRASSED AREAS SHALL BE RESODDED & RESTORED PRIOR TO PROJECT TURN–OVER. RESTORATION AND RESODDING SHALL BE CONSIDERED INCIDENTAL TO THE CONSTRUCTION OF REQUIRED IMPROVEMENTS.
10. ALL AREAS DISTURBED, DAMAGED, OR DEGRADED BY THE CONTRACTOR SHALL BE RESTORED TO PRE–CONSTRUCTION CONDITION OR BETTER.
11. IT IS THE CONTRACTOR'S RESPONSIBILITY TO COMPLY WITH ALL REQUIREMENTS OF THE FDEP NPDES PERMIT FOR STORMWATER DISCHARGE FROM LARGE & SMALL CONSTRUCTION ACTIVITIES AND THE STORMWATER POLLUTION PREVENTION PLAN (SWPPP). ALL DISCHARGE FROM DEWATERING OPERATIONS SHALL BE CONTAINED ON–SITE.
12. REFER TO LANDSCAPE PLANS FOR TREE PRESERVATION OR REMOVAL.

ALL ELEVATIONS REFER TO NORTH AMERICAN VERTICAL DATUM OF 1988 (NAVD 88). TO CONVERT TO NGVD 29 DATUM, ADD 1.54' TO NAVD ELEVATIONS.

NOTE: THESE PLANS ARE PRELIMINARY AND SHOULD NOT BE USED FOR CONSTRUCTION OR BIDDING/CONTRACTUAL PURPOSES UNTIL ALL PERMITS ARE ISSUED.

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FIELD:
DRAWN: P. Saffold
DESIGNED: TNM
APPROVED: TNM
PROJECT #21-030

NO.	DATE	REVISIONS
1	1/23/2022	REV PER TOWN

SITE DEMOLITION & EROSION CONTROL PLAN
1319 MANGONIA PARK WAREHOUSE
[redacted]
MANGONIA PARK, FLORIDA

TOWN RESUBMITTAL
DATE: 8/1/2024

SHEET
C5.1
OF 13

GENERAL NOTES:

1. COORDINATION OF DRAWINGS AND DISCIPLINES. IT IS THE CONTRACTOR'S RESPONSIBILITY TO REVIEW ALL DRAWINGS AND SPECIFICATIONS FOR COMPLETENESS AND ACCURACY. FURTHER CONTRACTOR IS COORDINATE ALL TRADES AND DISCIPLINES PROPERLY. ANY DISCREPANCIES IN PLANS OR SPECIFICATIONS ARE TO BE BROUGHT TO THE ATTENTION OF THE ENGINEER IN WRITING.

2. TOWN R/W. ALL MATERIALS AND CONSTRUCTION WITHIN THE TOWN RIGHT-OF-WAY SHALL CONFORM TO THE LATEST FDOT DESIGN STANDARDS & LATEST SPECIFICATIONS FOR ROAD & BRIDGE CONSTRUCTION.

3. BUILDINGS & UTILITIES. CONTRACTOR IS ADVISED THAT THE BUILDINGS AND ASSOCIATED SYSTEMS ARE STILL UNDER REVIEW BY VARIOUS JURISDICTIONAL AGENCIES. AS SUCH, THERE MAY BE MODIFICATIONS TO SERVICE LINES (MATERIALS, SIZES, LOCATIONS, DEPTHS, ETC.) AS THE DOCUMENTS ARE PERMITTED.

4. SAFETY. IT SHALL BE THE CONTRACTOR'S SOLE RESPONSIBILITY TO TAKE THE NECESSARY PRECAUTIONS TO ENSURE PROPER SAFETY AND WORKMANSHIP WHEN WORKING IN THE VICINITY OF EXISTING UTILITY LINES. THE CONTRACTOR SHALL BE RESPONSIBLE TO COORDINATE WITH THE RESPECTIVE UTILITY PROVIDER ON ANY WORK IN THE VICINITY OF OVERHEAD OR UNDERGROUND LINES. CONTRACTOR SHALL VERIFY PROPER CLEARANCE BELOW EXISTING OVERHEAD LINES PRIOR TO WORKING WITHIN THE VICINITY OF THE LINES.

5. FAMILIARITY WITH SITE CONDITIONS. THE CONTRACTOR SHALL VISIT THE SITE AND ACQUAINT HIMSELF WITH ALL EXISTING CONDITIONS. MAKE HIS OWN SUBSURFACE INVESTIGATION TO SATISFY HIMSELF AS TO THE SITE AND SUBSURFACE CONDITIONS, BUT SUCH SUBSURFACE INVESTIGATIONS SHALL BE PERFORMED ONLY UNDER TIME SCHEDULES AND ARRANGEMENTS APPROVED IN ADVANCE BY THE OWNER'S REPRESENTATIVE AND ENGINEER. THE EXISTING GRADES SHOWN ON THE DRAWINGS ARE APPROXIMATE ONLY AND NO REPRESENTATION IS MADE AS TO THEIR ACCURACY OR CONSISTENCY. THE CONTRACTOR SHALL VERIFY ALL EXISTING GRADES TO THE EXTENT NECESSARY TO INSURE COMPLETION OF THE JOB TO THE PROPOSED GRADES INDICATED ON THE DRAWINGS.

6. WORKMANSHIP. ALL WORK SHALL BE PERFORMED IN A WORKMANSHIP-LIKE MANNER AND SHALL CONFORM WITH ALL APPLICABLE CITY, COUNTY, STATE AND FEDERAL REGULATIONS AND/OR CODES, INCLUDING OSHA.

7. ELEVATIONS. ALL ELEVATIONS REFER TO THE NORTH AMERICAN VERTICAL DATUM OF 1988 (NAVD 88).

8. SURVEY & BENCHMARK. REFER TO SURVEY PREPARED BY PM SURVEYING FOR ADDITIONAL INFORMATION.

9. GEOTECHNICAL INFORMATION. REFER TO GEOTECHNICAL REPORTS AND BORING LOGS FOR ADDITIONAL INFORMATION INCLUDING SITE PREPARATION REQUIREMENTS.

10. FIELD LOCATIONS. THE CONTRACTOR SHALL CONTACT "SUNSHINE STATE ONE CALL OF FLORIDA, INC. (1-800-432-4770)" AT LEAST 48 HOURS PRIOR TO BEGINNING ANY EXCAVATION. THE CONTRACTOR SHALL GIVE ADEQUATE NOTIFICATION TO ALL AFFECTED UTILITY OWNERS FOR REMOVAL, RELOCATION AND ALTERATION OF THEIR EXISTING FACILITIES.

11. PROTECTION OF UTILITIES. THE CONTRACTOR SHALL BE RESPONSIBLE AT ALL TIMES THROUGHOUT THE DURATION OF CONSTRUCTION FOR THE PROTECTION OF ALL EXISTING AND NEWLY INSTALLED ABOVE GROUND, UNDERGROUND, AND ON THE SURFACE STRUCTURES AND UTILITIES FROM DAMAGE OR SERVICE DISRUPTION. THE CONTRACTOR SHALL BE RESPONSIBLE FOR TAKING SUCH MEASURES AS NECESSARY TO PROTECT THE HEALTH, SAFETY, AND WELFARE OF THOSE PERSONS HAVING ACCESS TO THE WORK SITE.

12. EXISTING UTILITIES. EXISTING UTILITY LOCATIONS SHOWN ON THESE PLANS ARE APPROXIMATE. THE ENGINEER ASSUMES NO RESPONSIBILITY FOR THE ACCURACY OF THE EXISTING UTILITIES SHOWN ON THE PLANS OR FOR ANY EXISTING UTILITIES THAT ARE NOT SHOWN ON THE PLANS. PRIOR TO THE START OF ANY CONSTRUCTION ACTIVITY, IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO MAKE ARRANGEMENTS FOR FIELD LOCATIONS OF EXISTING UTILITIES. ANY RELOCATIONS OF THE VARIOUS EXISTING UTILITIES SHALL BE COORDINATED WITH THE RESPECTIVE UTILITY PROVIDER IN A TIMELY MANNER TO MINIMIZE IMPACT ON THE CONSTRUCTION SCHEDULE. ANY DELAY CAUSED DUE TO THE RELOCATION OF UTILITIES SHALL BE INCIDENTAL TO THE CONTRACT AND NO EXTRA COMPENSATION WILL BE ALLOWED. ALL COSTS ASSOCIATED WITH VERIFICATION, PROTECTION OF EXISTING SYSTEMS OR CONNECTION TO THOSE SYSTEMS AS IDENTIFIED ON THE PLANS IS INCLUDED IN THE CONTRACTOR'S BID.

13. EXISTING UTILITIES. PRIOR TO BEGINNING CONSTRUCTION, THE CONTRACTOR SHALL VERIFY THE SIZE, LOCATION, ELEVATION, AND MATERIAL OF ALL EXISTING UTILITIES WITHIN THE AREA OF CONSTRUCTION. THE CONTRACTOR IS RESPONSIBLE FOR VERIFICATION OF ALL EXISTING UTILITIES AND SOILS CONDITIONS AND SHALL NOTIFY THE ENGINEER IN THE EVENT OF ANY DISCREPANCIES WITH THE PLANS.

14. EXISTING UTILITIES. THE CONTRACTOR SHALL BE RESPONSIBLE FOR DAMAGE TO ANY EXISTING UTILITIES FOR WHICH IT FAILS TO REQUEST LOCATIONS FROM THE UTILITY OWNER. THE CONTRACTOR IS RESPONSIBLE AS WELL FOR DAMAGE TO ANY EXISTING UTILITIES WHICH ARE PROPERLY LOCATED.

15. FIELD REVIEWS. THE CONTRACTOR SHALL NOTIFY THE ENGINEER 48 HOURS PRIOR TO ANY REQUIRED FIELD REVIEWS OR INSPECTIONS AND SHALL SUPPLY ALL EQUIPMENT NECESSARY, LABOR AND MATERIALS FOR INSPECTION AND/OR TEST. ALL WORK SHALL BE OPEN AND SUBJECT TO REVIEW AND/OR INSPECTION BY AUTHORIZED PERSONNEL OF THE CITY, OWNER, INVOLVED UTILITY COMPANIES, PROJECT ENGINEER, AND REGULATORY AGENCIES. ALL RECOMMENDATIONS AND REQUIREMENTS OF INSPECTION PERSONNEL OTHER THAN THE OWNER'S SHALL BE REPORTED TO THE ENGINEER PRIOR TO IMPLEMENTATION. COMPENSATION WILL NOT BE ALLOWED FOR WORK WHICH IS NOT AUTHORIZED BY THE ENGINEER OR OWNER.

16. DIFFERING SITE CONDITIONS. THE CONTRACTOR SHALL IMMEDIATELY NOTIFY THE ENGINEER OF ANY DISCREPANCIES FOUND BETWEEN THE DRAWINGS AND THE FIELD CONDITIONS PRIOR TO CONSTRUCTION IN THE AREA OF CONFLICT. ANY DIFFERING SITE CONDITIONS FROM THAT WHICH ARE REPRESENTED ON THE PLANS, WHETHER ABOVE, ON OR BELOW THE SURFACE OF THE GROUND, ARE TO BE BROUGHT TO THE IMMEDIATE ATTENTION OF THE ENGINEER AND THE OWNER IN WRITING.

17. REQUIRED PERMITS. IT IS THE RESPONSIBILITY OF THE CONTRACTOR(S) TO ENSURE THAT ALL REQUIRED PERMITS ARE OBTAINED AND AT THE JOB SITE PRIOR TO THE COMMENCEMENT OF CONSTRUCTION. CONTRACTOR SHALL ABIDE BY ALL PERMIT CONDITIONS. PERMITS AND APPROVALS APPLICABLE TO THIS PROJECT INCLUDE:

   A. TOWN OF MANGONIA PARK
   B. SOUTH FLORIDA WATER MANAGEMENT DISTRICT
   C. FLORIDA DEPARTMENT OF ENVIRONMENTAL PROTECTION NPDES

19. DEWATERING. THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING AND COMPLYING WITH ANY DEWATERING PERMITS AND/OR APPROVALS NECESSARY FOR THE CONSTRUCTION OF THE PROPOSED FACILITIES AND SYSTEMS.

20. PERMITS. THE CONTRACTOR SHALL OBTAIN ALL REQUIRED PERMITS AND/OR LICENSES TO COMMENCE CONSTRUCTION.

21. USE OF EXISTING MATERIALS. NO EXISTING MATERIAL SHALL BE USED IN NEW CONSTRUCTION UNLESS APPROVED DURING THE SHOP DRAWING APPROVAL PROCESS.

22. EXISTING LANDSCAPE. CONTRACTOR SHALL BE EXTREMELY CAUTIOUS WHEN WORKING NEAR TREES WHICH ARE TO BE SAVED, WHETHER SHOWN ON THE PLAN OR DESIGNATED IN THE FIELD.

23. SHOP DRAWINGS. THE CONTRACTOR SHALL PROVIDE A MINIMUM OF THREE (3) COPIES (OR MORE IF SPECIFIED IN THE CONTRACT) OF SHOP DRAWINGS, PRODUCT DATA, MATERIAL SPECIFICATIONS AND OTHER INFORMATION REGARDING CONSTRUCTION MATERIALS AND STRUCTURES AS REQUESTED BY THE OWNER, ENGINEER OR JURISDICTIONAL AGENCY. SHOP DRAWINGS MUST BE NEWLY PREPARED INFORMATION, DRAWN TO ACCURATE SCALE. DO NOT REPRODUCE CONTRACT DOCUMENTS OR COPY STANDARD INFORMATION AS THE BASIS OF SHOP DRAWINGS. STANDARD INFORMATION PREPARED WITHOUT SPECIFIC REFERENCE TO THE PROJECT WILL NOT BE CONSIDERED A SHOP DRAWING.

   A. BY SUBMITTAL OF ANY SHOP DRAWING OR CATALOG DATA, BEARING AN APPROVAL STAMP, THE CONTRACTOR REPRESENTS THAT IT HAS DETERMINED AND VERIFIED ALL FIELD MEASUREMENTS, FIELD CONSTRUCTION CRITERIA, MATERIALS, DIMENSIONS, CATALOG NUMBERS AND SIMILAR DATA, OR WILL DO SO, AND THAT IT HAS CHECKED AND COORDINATED EACH ITEM WITH OTHER APPLICABLE APPROVED SHOP DRAWINGS AND THE CONTRACT REQUIREMENTS.

   B. SHOP DRAWINGS AND CATALOG DATA SUBMITTED WITHOUT THE CONTRACTOR'S STAMP OF APPROVAL WILL BE RETURNED TO THE CONTRACTOR WITHOUT REVIEW. APPROVAL OF SHOP DRAWINGS, SAMPLES, OR CATALOG DATA BY THE ENGINEER SHALL NOT AUTHORIZE ANY DEVIATION FROM THE REQUIREMENT OF THE CONTRACT DOCUMENTS.

   C. ANY PROPOSED SUBSTITUTE OR EQUAL TO THE THAT SHOWN ON THE CONTRACT DOCUMENTS SHALL BE ACCOMPANIED BY CALCULATIONS SUBSTANTIATING EQUIVALENCY. SHOP DRAWINGS WITH SUBSTITUTE MATERIALS NOT ACCOMPANIED BY CALCULATIONS WILL BE RETURNED WITHOUT REVIEW.

   D. THE CONTRACTOR SHALL CONFIRM COMPATIBILITY OF PIPE SLOPES AND INVERTS DURING SHOP DRAWING AND MATERIALS ORDERING PHASE OF THE PROJECT AND ADVISE THE ENGINEER IN WRITING OF ANY DISCREPANCIES.

   E. INDIVIDUAL SHOP DRAWINGS FOR ALL PRECAST STRUCTURES ARE REQUIRED. CATALOG LITERATURE WILL NOT BE ACCEPTED FOR PRECAST STRUCTURES.

24. TESTING. THE CONTRACTOR SHALL BE RESPONSIBLE FOR ALL REQUIRED TESTS TO BE PERFORMED BY AN INDEPENDENT TESTING LABORATORY APPROVED BY THE ENGINEER. THE CONTRACTOR UNDERSTANDS THAT ANY AND ALL TESTS, INSPECTIONS, PROCEDURES INCLUDING ALL LABOR, MATERIALS, TIME AND EFFORT IN ORDER TO HAVE ALL WORK DESIGNATED AS FINAL BY THE RESPECTIVE JURISDICTIONAL AUTHORITIES AND THAT SUCH TESTS, INSPECTIONS AND PROCEDURES ARE INCLUDED IN THE BID.

25. RECORD DRAWINGS. RECORD DRAWINGS SHALL BE SUBMITTED IN SUFFICIENT QUANTITIES AS REQUIRED BY THE OWNER, ENGINEER AND JURISDICTIONAL AGENCIES TO COMPLETE THE CLOSE-OUT OF THE PROJECT.

   A. THE CONTRACTOR SHALL PROVIDE THE ENGINEER OF RECORD WITH ADEQUATE RECORD DRAWING INFORMATION, WHICH HAS BEEN PREPARED BY A LAND SURVEYOR REGISTERED IN THE STATE OF FLORIDA. THE INFORMATION SHALL BE PROVIDED ON A 24"X36" REPRODUCIBLE MYLAR. THE SURVEYOR SHALL SIGN AND SEAL THE RECORD DRAWINGS. THE FOLLOWING SHALL BE THE MINIMUM REQUIREMENTS FOR RECORD DRAWING INFORMATION:

   B. THE WORDS "RECORD DRAWINGS" SHALL BE WRITTEN IN BOLD LETTERS IN A CONSPICUOUS PLACE. NO RUBBER STAMP ON PLANS OR MYLARS SHALL BE ACCEPTED.

   C. ANY SIGNIFICANT CHANGES IN ALIGNMENT, PIPE CONFLICTS, LOCATION OF STORM MANHOLES, CATCH BASINS, ETC. SHALL BE SHOWN TO SCALE ON RECORD DRAWINGS. THE "PROPOSED" ALIGNMENT, ETC. SHALL BE CROSSED OUT.

   D. THE DRAWINGS SHALL BE LEGIBLE AND THE AS-BUILT DIMENSIONS EASY TO IDENTIFY.

   E. DATA FOR STORM MANHOLES AND CATCH BASINS (RIM, INVERTS), PIPE SLOPES, AND PIPE LENGTHS AS CONSTRUCTED SHALL BE SHOWN ON THE PLAN SHEET NEXT TO "PROPOSED" DIMENSIONS. THE "PROPOSED" NUMBERS SHALL BE CROSSED OUT. IF THE ASBUILT CONDITION IS THE SAME AS THE PROPOSED INFORMATION, THEN THE PROPOSED INFORMATION WILL BE CROSSED OUT AND THE ASBUILT DATA WRITTEN.

   F. RECORD DRAWING INFORMATION SHALL BE PROVIDED FOR THE CONTROL STRUCTURE, INCLUDING ALL PIPE INVERTS AND PIPE SIZES, AND GRATE ELEVATION. ELEVATIONS AND DIMENSIONS SHALL BE PROVIDED ON THE WEIR, THE BLEEDER ORIFICE AND THE POLLUTION CONTROL BAFFLE (IF APPLICABLE). THE BOTTOM ELEVATION OF STRUCTURE AND ELEVATIONS AND DIMENSIONS OF ANY SLOTS OR ORIFICES IN THE INLET WALLS SHALL BE PROVIDED.

   G. PERIMETER BERM ELEVATIONS ADJACENT TO THE PROPERTY LINES SHALL BE PROVIDED EVERY 200 LINEAR FEET MINIMUM.

   H. CROSS SECTIONS OF DRY DETENTION AREAS AND WATER MANAGEMENT LAKE SECTIONS SHALL BE TAKEN EVERY 200 LINEAR FEET MINIMUM. SECTIONS ON WATER MANAGEMENT LAKES SHALL INCLUDE THE FROM THE UPPER ELEVATION OF THE 20 FOOT LAKE MAINTENANCE AREA TO 2 FEET BELOW CONTROL WATER ELEVATION. CROSS SECTIONS SHALL DEPICT THE PROPOSED AND ASBUILT DATA INCLUDING ALL GRADE BREAKS.

26. FINAL UTILITY ADJUSTMENTS. ALL UTILITIES SHALL BE ADJUSTED TO NEW FINISH GRADES AND PROPERLY SET TO PAVEMENT CROSS SLOPE AS REQUIRED.

27. DETECTABLE WARNING. CONTRACTOR SHALL CONFIRM DETECTABLE WARNING REQUIREMENTS WITH BUILDING OFFICIAL PRIOR TO INSTALLATION.

SITE DEMOLITION AND EROSION CONTROL NOTES:

1. SITE ACCESS. SUITABLE STABILIZED ROADS ARE TO BE PROVIDED DURING CONSTRUCTION FOR FIRE AND PARAMEDICAL ACCESS.

2. SITE. CONTRACTOR ACCEPTS THE SITE AS IS. ANY REPAIRS TO EXISTING SITE ELEMENTS INCLUDING PAVEMENT, SIDEWALKS, CURBING, DRAINAGE, SANITARY SEWER, WATER, SIGNAGE & STRIPING SHALL BE INCLUDED IN THE BID PRICE.

3. EROSION CONTROL. THE CONTRACTOR SHALL INSTALL AND MAINTAIN EROSION AND TURBIDITY BARRIERS PRIOR TO AND DURING CONSTRUCTION TO PREVENT VIOLATION OF STATE WATER QUALITY STANDARDS FOR OFFSITE DISCHARGES THESE BARRIERS SHALL REMAIN IN PLACE UNTIL CONSTRUCTION IS COMPLETED, SOILS ARE STABILIZED, AND VEGETATION HAS BEEN ESTABLISHED. REFERENCE FDOT ROADWAY AND TRAFFIC DESIGN STANDARDS INDICES 102, 103 AND 106 FOR TYPICAL INSTALLATION DETAILS.

4. FILTER FABRIC. FILTER FABRIC SHALL BE INSTALLED ON ALL CATCH BASIN GRATES AND MAINTAINED UNTIL CONSTRUCTION IS COMPLETE.

5. ADJACENT IMPACTS. CONTRACTOR SHALL PROTECT ADJACENT WATER BODIES, WETLANDS AND PROPERTIES FROM DAMAGE BY SEDIMENTATION OR OTHER POTENTIAL CONSTRUCTION RELATED CAUSES.

6. SILT FENCES. SILT FENCES SHALL BE INSTALLED AND MAINTAINED AROUND ALL CLEARING PERIMETERS.

7. SITE CLEARING. CLEAR AREAS INDICATED SHALL BE COMPLETELY CLEAR OF ALL TIMBER, BRUSH, STUMPS, ROOTS, GRASS, WEEDS, RUBBISH AND ALL OTHER DEBRIS AND OBSTRUCTIONS RESTING ON OR PROTRUDING THROUGH THE SURFACE OF THE GROUND. THIS DEBRIS SHALL BE DISPOSED OF IN A LEGAL MANNER. BURNING OF THIS MATERIAL IS NOT PERMITTED UNLESS THE CONTRACTOR OBTAINS SPECIFIC PERMITS ALLOWING SUCH ACTIVITY.

GEOMETRIC, SIGNING, AND STRIPING NOTES:

1. REFERENCES. WORK SHALL BE PERFORMED IN ACCORDANCE WITH THE FLORIDA DEPARTMENT OF TRANSPORTATION (FDOT) STANDARD SPECIFICATIONS FOR ROAD AND BRIDGE CONSTRUCTION (LATEST EDITION), FDOT TRAFFIC DESIGN STANDARDS (LATEST EDITION), MANUAL ON UNIFORM TRAFFIC CONTROL DEVICES (MUTCD) (LATEST EDITION) AND THE PALM BEACH COUNTY TYPICALS FOR PAVEMENT MARKINGS, SIGNING & GEOMETRICS (T-P-13-001). ALL REFERENCES ARE INCORPORATED INTO THE PLAN DOCUMENTS. CONTRACTOR IS RESPONSIBLE FOR OBTAINING COMPLETE COPIES OF THE LATEST EDITION OF THE REFERENCES.

2. PARKING LOT STRIPING. ALL PARKING STALL STRIPING SHALL BE TRAFFIC PAINT AND CONFORM TO THE RESPECTIVE JURISDICTIONAL AUTHORITY REQUIREMENTS. ALL OTHER PARKING LOT STRIPING SHALL BE THERMOPLASTIC PER FDOT SPECIFICATIONS FOR ROAD & BRIDGE CONSTRUCTION.

3. ROADWAY STRIPING. ALL ROADWAY STRIPING SHALL BE THERMOPLASTIC IN ACCORDANCE WITH THE RESPECTIVE JURISDICTIONAL AUTHORITY REQUIREMENTS.

4. BUILDING TIES AND DIMENSIONS. BUILDING TIES SHOWN ARE APPROXIMATE. CONTRACTOR TO VERIFY WITH ARCHITECTURAL PLANS. BUILDINGS SHOWN ON THESE DRAWINGS MAY NOT REFLECT MINOR ARCHITECTURAL REVISIONS AND SHOULD NOT BE RELIED UPON FOR STAKING OF BUILDINGS. CONTRACTOR SHOULD OBTAIN LATEST ARCHITECTURAL INFORMATION AND ESTABLISH GEOMETRIC DATA FOR STAKING. SHOULD THE CONTRACTOR IDENTIFY A CONFLICT BETWEEN THE ARCHITECTURAL INFORMATION AND THESE DRAWINGS, NOTIFY THE ENGINEER IN WRITING.

5. MAINTENANCE OF TRAFFIC. CONTRACTOR SHALL BE RESPONSIBLE FOR FURNISHING A PROPER TRAFFIC MAINTENANCE AND CONTROL PLAN IN ACCORDANCE WITH REGULATORY STANDARDS. INCLUDED WILL BE THE LABOR, EQUIPMENT AND MATERIALS REQUIRED TO IMPLEMENT THAT PLAN. ALL WORK MUST BE DONE IN ACCORDANCE WITH AGENCY REQUIREMENTS AND ANY REQUIRED PERMITS OR APPROVALS MUST BE APPLIED FOR AND OBTAINED BY THE CONTRACTOR.

ALL ELEVATIONS REFER TO NORTH AMERICAN VERTICAL DATUM OF 1988 (NAVD 88).
TO CONVERT TO NGVD 29 DATUM, ADD 1.54' TO NAVD ELEVATIONS.

NOTE: THESE PLANS ARE PRELIMINARY AND SHOULD NOT BE USED FOR CONSTRUCTION OR BIDDING/CONTRACTUAL PURPOSES UNTIL ALL PERMITS ARE ISSUED.

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FIELD:
DRAWN: P. Saffold
DESIGNED: TNM
APPROVED: TNM

PROJECT #21-030

NO.	DATE	REVISIONS

GENERAL NOTES & SPECIFICATIONS

1319 MANGONIA PARK WAREHOUSE

[redacted]
MANGONIA PARK, FLORIDA

TOWN RESUBMITTAL
DATE: 8/1/2024

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PAVING, GRADING, AND DRAINAGE NOTES:

1. REFERENCES. WORK SHALL BE PERFORMED IN ACCORDANCE WITH THE FLORIDA DEPARTMENT OF TRANSPORTATION (FDOT) STANDARD SPECIFICATIONS FOR ROAD AND BRIDGE CONSTRUCTION, LATEST EDITION, AND FDOT ROADWAY TRAFFIC DESIGN STANDARDS, LATEST EDITION. ALL FDOT INDICES ARE INCORPORATED AS PLAN REFERENCES. CONTRACTOR IS RESPONSIBLE FOR OBTAINING COMPLETE COPIES OF THE LATEST EDITION OF THE FDOT INDICES.

2. SITE FAMILIARIZATION. PRIOR TO BID PREPARATION, THE CONTRACTOR MUST BECOME FAMILIAR WITH THE OVERALL SITE CONDITIONS AND PERFORM ADDITIONAL INVESTIGATIONS AS DETERMINED NECESSARY TO UNDERSTAND THE LIMIT AND DEPTH OF EXPECTED ORGANIC MATERIAL, ADEQUACY OF EXISTING MATERIALS AS FILL, DEWATERING REQUIREMENTS, CLEAN FILL REQUIRED FROM OFFSITE AND MATERIALS TO BE DISPOSED OF OFFSITE. ANY DELAY, INCONVENIENCE OR EXPENSE CAUSED TO THE CONTRACTOR DUE TO INADEQUATE INVESTIGATION OF EXISTING CONDITIONS SHALL BE INCIDENTAL TO THE CONTRACT AND NO EXTRA COMPENSATION WILL BE ALLOWED. THE MATERIALS ANTICIPATED TO BE ENCOUNTERED DURING CONSTRUCTION MAY REQUIRE DRYING PRIOR TO USE AS BACKFILL AND THE CONTRACTOR MAY HAVE TO IMPORT MATERIALS, AT NO EXTRA COST, FROM OFFSITE TO MEET THE REQUIREMENTS FOR COMPACTION AND PROPER FILL.

3. SITE FILL BALANCE. IT IS THE CONTRACTOR'S RESPONSIBILITY TO DETERMINE EXISTING SITE CONDITIONS OF SOIL DURING THE BID PREPARATION TO DETERMINE IF ANY OFF SITE MATERIALS WILL NEED TO BE IMPORTED TO ACHIEVE THE GRADES SPECIFIED ON THE PLANS.

4. RESTORATION. ALL DISTURBED AREAS SHALL BE RESTORED TO PRE-CONSTRUCTION CONDITION OR BETTER AS DETERMINED BY THE ENGINEER, OWNER AND RESPECTIVE JURISDICTIONAL AUTHORITY.

5. HORIZONTAL CONTROL AND LAYOUT. CONTRACTOR/OWNER SHALL COORDINATE WITH A REGISTERED SURVEYOR TO LAY OUT LINE, GRADES, STRUCTURES, LOT CORNERS, ETC. AS NECESSARY TO CONSTRUCT THE PROPOSED FACILITIES IN ACCORDANCE WITH THE PLANS AND SPECIFICATIONS. BENCH MARKS SHALL BE ESTABLISHED BY A REGISTERED SURVEYOR. BENCH MARKS SHALL BE PERMANENT AND EASILY ACCESSIBLE AND MAINTAINED AND REPLACED IF DISTURBED OR DESTROYED. HORIZONTAL CONTROL SHALL SIMILARLY BE ESTABLISHED AND MAINTAINED.

6. CLEARING, GRADING AND DISPOSAL. CLEARING AND GRADING SHALL CONFORM TO SECTION 110 OF THE FDOT STANDARD SPECIFICATIONS FOR ROAD AND BRIDGE CONSTRUCTION, LATEST EDITION. CLEARING SHALL BE CONTAINED TO THE CONSTRUCTION AREA AND/OR AS DIRECTED BY THE OWNER AND/OR HIS REPRESENTATIVE. ALL ROOTS, STUMPS, OR OTHER OBSTRUCTIONS SHALL BE REMOVED TO A DEPTH OF 18" BELOW NATURAL GRADE OR DESIGN GRADE WHICHEVER IS LOWER. ALL DEBRIS SHALL BE REMOVED FROM THE AREA AND BE DISPOSED OF LEGALLY.

7. EXCAVATION AND EMBANKMENT. ALL EXCAVATION AND EMBANKMENT SHALL CONFORM TO THE FDOT STANDARD SPECIFICATIONS FOR ROAD AND BRIDGE CONSTRUCTION, LATEST EDITION.

8. EARTHWORK. NO SEPARATE PAYMENT SHALL BE PROVIDED TO THE CONTRACTOR FOR EARTHWORK ASSOCIATED WITH THIS CONTRACT. ALL COSTS ASSOCIATED WITH SITE PREPARATION NECESSARY TO EXECUTE THE WORK SET FORTH IN THE CONTRACT SHALL BE INCLUDED IN THE BID.

9. EXCESS MATERIAL. ANY EXCESS MATERIAL REMAINING UPON COMPLETION OF CONSTRUCTION SHALL BE REMOVED TO AN ACCEPTABLE LOCATION AT THE EXPENSE OF THE CONTRACTOR. AT THE OWNER'S OPTION, A LOCATION MAY BE IDENTIFIED FOR MATERIAL STOCK PILING OR THE OWNER MAY INSTRUCT THE CONTRACTOR TO REMOVE THE EXCESS MATERIAL IN WHICH CASE THE CONTRACTOR IS TO LEGALLY DISPOSE OF ALL MATERIALS OFFSITE.

10. FINISHED SITE GRADING. THE CONTRACTOR SHALL BE RESPONSIBLE FOR FINAL GRADING OF THE SITE TO WITHIN ±0.1 FOOT OF ALL THE PROPOSED FINISHED GRADE ELEVATIONS THROUGHOUT THE PROJECT SITE AS DEPICTED ON THE PLANS. ELEVATIONS OF CONSTRUCTED FACILITIES, I.E. ROADWAYS, DRAINAGE, ETC. ARE TO BE AS INDICATED ON THE PLANS UNLESS APPROVED DIFFERENTLY IN WRITING.

11. BUILDING AREAS. ALL ORGANIC AND OTHER UNSUITABLE MATERIAL IN BUILDING OR PAVEMENT AREAS SHALL BE REMOVED AND REPLACED IN ACCORDANCE WITH THE GEOTECHNICAL REPORT RECOMMENDATIONS AND SOUND CONSTRUCTION PRACTICES.

12. LANDSCAPE AREAS. ALL LANDSCAPE ISLANDS SHALL CONTAIN SUITABLE LANDSCAPE MATERIAL FOR LANDSCAPE INSTALLATION. ALL LANDSCAPE AREAS SHALL BE SUITABLY SLOPED TO PROVIDE POSITIVE DRAINAGE.

13. SOD AREAS. ELEVATIONS OF GRASSED AREAS ARE GIVEN AT FINISHED SOD/SEED GRADE. CONTRACTOR TO GRADE AREAS RECEIVING SOD LOWER TO ACCOMMODATE SOD THICKNESS.

14. UNSUITABLE MATERIALS. WHERE ENCOUNTERED, UNSUITABLE MATERIAL SHALL BE REMOVED TO A DEPTH AND AREA DETERMINED BY THE ENGINEER. BACKFILLING SHALL BE IN LAYERS NOT GREATER THAN 8" THICKNESS AND COMPACTED TO 98 PERCENT OF MAXIMUM DENSITY AS DETERMINED BY AASHTO T-180. ANY UNSTABLE SOIL AND/OR SOIL CONTAINING ORGANIC MATERIAL ENCOUNTERED DURING CONSTRUCTION, INCLUDING BET NOT LIMITED TO MUCK, SILT, SLUDGE, AND MARL SHALL BE REMOVED AND SHALL BE REPLACED BY MATERIAL ACCEPTABLE TO THE ENGINEER. IF ANY ROCK, HARDPAN, ETC. IS ENCOUNTERED DURING CONSTRUCTION OF ANY ITEMS, THE REMOVAL AND BACKFILL REQUIRED SHALL BE CONSIDERED INCIDENTAL TO CONSTRUCTION OF THAT ITEM AND NO SEPARATE PAYMENT WILL BE MADE.

15. GRADING. GRADING FROM PROPOSED TO EXISTING CONDITIONS SHALL NOT BE STEEPER THAN 3H:1V NOR FLATTER THAN 20H:1V. ALL SWALES AND SLOPES SHALL BE SODDED AFTER GRADING.

16. ELEVATIONS. ELEVATIONS GIVEN ADJACENT TO CURB OR FRONT OF SIDEWALK ARE CONSIDERED EDGE OF PAVEMENT GRADES UNLESS OTHERWISE NOTED.

17. POSITIVE DRAINAGE. CONTRACTOR IS RESPONSIBLE FOR GRADING ALL PAVEMENT TO DRAIN POSITIVELY. INTERSECTIONS SHALL BE TRANSITIONED TO PROVIDE A SMOOTH DRIVING SURFACE WHILE MAINTAINING POSITIVE DRAINAGE. SHOULD AREAS OF POOR DRAINAGE BE OBSERVED, THE CONTRACTOR SHALL NOTIFY THE ENGINEER PRIOR TO PLACEMENT OF CURBS OR PAVEMENT COURSES, SO THAT RECOMMENDATIONS FOR CORRECTION CAN BE MADE.

18. ADA AREAS. PAVEMENT GRADES AT HANDICAP PARKING SPACES SHALL NOT EXCEED 2% IN ANY DIRECTION. PROPOSED AND EXISTING SIDEWALKS SHALL BE RAMPED FLUSH WITH PAVEMENT. RAMPS SHALL NOT EXCEED A 12H:1V SLOPE. SIDEWALK RAMPS SHALL BE TEXTURED IN CONFORMANCE WITH FDOT STANDARDS EXCEPT WHERE LOCAL CODES DICTATE OTHERWISE OR IF INDICATED OTHERWISE ON THE PLANS. CONTRACTOR SHALL COORDINATE WITH THE ENGINEER AND THE JURISDICTIONAL AUTHORITY FOR FORM BOARD AND PRE-POUR INSPECTIONS PRIOR TO ANY CONCRETE INSTALLATION. IT IS THE CONTRACTOR'S SOLE RESPONSIBILITY TO ENSURE THAT ALL RAMP TEXTURED SURFACES AND SIDEWALK LONGITUDINAL AND CROSS SLOPES ARE IN CONFORMANCE WITH LOCAL, STATE AND FEDERAL ADA AND BUILDING CODE STANDARDS.

19. ROADWAY CONSTRUCTION. ALL ROADWAY CONSTRUCTION AND MATERIAL SHALL CONFORM TO THE APPLICABLE FDOT STANDARD SPECIFICATIONS FOR ROAD AND BRIDGE CONSTRUCTION, LATEST EDITION. ALL SUBBASE, BASE COURSE, PRIME COAT, TACK COAT AND ASPHALT MATERIALS SHALL BE INSTALLED IN ACCORDANCE WITH FDOT STANDARD SPECIFICATIONS (LATEST EDITION).

20. DRAINAGE STRUCTURES. ALL DRAINAGE STRUCTURES SHALL BE CONSTRUCTED AS PER THE PLANS AND IN ACCORDANCE WITH FDOT STANDARD SPECIFICATIONS FOR ROAD AND BRIDGE CONSTRUCTION (LATEST EDITION). ALL GRATES SHALL BE SECURED TO THE DRAINAGE STRUCTURE WITH A CHAIN AND EYEBOLT.

21. PIPE LENGTHS. PIPE LENGTHS SHOWN ON THE PLANS REPRESENT SCALED DISTANCES BETWEEN CENTERLINES OF DRAINAGE STRUCTURES AND FROM INVERTS OF ENDWALLS AND/OR MITERED END SECTIONS.

22. PIPE INSTALLATION. PIPE TRENCH SHALL BE DRY WHILE PIPE IS BEING LAID AND TO BE BEDDED PER DETAIL AND IN CONFORMANCE WITH THE FDOT STANDARD SPECIFICATIONS FOR ROAD AND BRIDGE CONSTRUCTION (LATEST EDITION). ALL CORRUGATED PIES SHALL BE INSTALLED WITH MAXIMUM LENGTHS IN ORDER TO MINIMIZE JOINTS.

23. PIPE TRENCHES. PIPE TRENCHES TO BE EXCAVATED AND BACKFILLED IN ACCORDANCE WITH FDOT STANDARD SPECIFICATIONS FOR ROAD AND BRIDGE CONSTRUCTION (LATEST EDITION).

24. REINFORCED CONCRETE PIPE (RCP). IF REINFORCED CONCRETE PIPE (RCP) IS USED, IT SHALL BE PER ASTM C-76 CLASS III, UNLESS OTHERWISE SPECIFIED. INSTALLATION OF RCP SHALL BE IN ACCORDANCE WITH FDOT SPECIFICATIONS (LATEST EDITION).

25. CORRUGATED ALUMINUM PIPE (CAP). CAP SHALL CONFORM WITH THE REQUIREMENTS OF AASHTO M-196 AND WITH THE FDOT SPECIFICATIONS (LATEST EDITION). CAP TO BE HELICAL (SPIRAL FLOW). ALL CONNECTION BANDS, FOR CORRUGATED PIPES, SHALL BE CORRUGATED, LAP-TYPE, IN CONFORMANCE WITH AASHTO M-16, AND TO BE WATERTIGHT BY USING A GASKET AT LEAST SIX (6) INCHES WIDE. ALL JOINTS ON STORM SEWER PIPE SHALL BE MADE UP WITH ONE-HALF INCH NEOPRENE, ALL BANDS SHALL HAVE THE SAME CORRUGATION DESIGN AS THE PIPE. THE WIDTH OF THE BANDS SHALL BE AS FOLLOWS: 12 INCH WIDE UP TO 48 INCH DIAMETER PIPE AND 24 INCH WIDE OVER 48 INCH DIAMETER PIPE. THE REQUIREMENTS OF FDOT DM06036 WILL BE ADHERED TO WITH ANNULAR BANDING OF JOINTS REQUIRED.

26. ROOF DRAIN PIPING. ALL ROOF DRAIN PIPING (OUTSIDE THE BUILDING) TO BE PVC SCHEDULE 40 UNLESS OTHERWISE NOTED. ALL FITTINGS REQUIRED TO MAKE A FULLY OPERATIONAL DRAINAGE SYSTEM MAY NOT BE SHOWN ON THE PLANS. ALL FITTINGS AND MATERIALS REQUIRED TO COMPLETE THE PROJECT ARE TO BE INCLUDED IN THE CONTRACTOR'S BID.

27. DRAINAGE CONNECTIONS. CONTRACTOR SHALL BE RESPONSIBLE FOR REMOVAL OF DEBRIS AND SILT WHERE CONNECTIONS TO EXISTING DRAINAGE STRUCTURES OR SWALES ARE TO BE PERFORMED. SWALES, DITCHES, LAKES, ETC. SHALL BE RESTORED TO THEIR ORIGINAL DESIGN SECTION AND SODDED. SUCH REMOVAL AND RESTORATION SHALL BE TO THE POINT OF LEGAL POSITIVE OUTFALL.

28. DRAINAGE JOINT WRAPPING. ALL DRAINAGE PIPE JOINTS TO BE WRAPPED IN FILTER FABRIC PER FDOT INDEX 280. THE FIRST JOINT OF PIPE ENTERING OR LEAVING A DRAINAGE STRUCTURE SHALL BE WRAPPED WITH FILTER FABRIC PER FDOT SPECIFICATIONS.

29. DRAINAGE LAMPING. ALL DRAINAGE SYSTEMS SHALL BE PUMPED DOWN BELOW ONE-THIRD OF THE DIAMETER OF THE PIPE (FROM THE INVERT) AND LAMPED AS A REQUIREMENT OF THE FINAL DRAINAGE INSPECTION.

30. SUBGRADE. SUBBASE SHALL BE FREE OF MUCK, STUMPS, ROOTS, UNDERBRUSH, VEGETATIVE MATTER, GARBAGE, TRASH OR ANY MATERIAL THAT WILL NOT PROVIDE A SUITABLE STABLE, CLEAN SUBGRADE. SUBGRADE SHALL BE AS SHOWN ON THE PLANS AND TO BE COMPACTED TO 98 PERCENT OF THE MAXIMUM DENSITY AS DETERMINED BY AASHTO T-180 UNLESS OTHERWISE NOTED ON THE PLANS.

31. UNDER PAVEMENT UTILITIES AND CONDUITS. THE SEQUENCE OF CONSTRUCTION SHALL BE SUCH THAT ALL UNDERGROUND INSTALLATIONS OF ANY KIND, INCLUDING IRRIGATION AND ELECTRICAL CONDUITS, THAT WILL REMAIN UNDER THE PROPOSED PAVEMENT OR WITHIN 10 FEET OF ITS EDGES SHALL BE INSTALLED PRIOR TO CONSTRUCTION OF THE BASE.

32. BASE. BASE SHALL BE AS SHOWN ON SECTION AND SHALL BE COMPACTED TO 98 PERCENT OF THE MAXIMUM DENSITY AS DETERMINED BY AASHTO T-180 UNLESS OTHERWISE NOTED ON THE PLANS.

33. ASPHALT TACK AND PRIME COATS. PRIME COAT SHALL BE APPLIED AT THE RATE OF 0.15 GALLONS PER SQUARE YARD. WHERE REQUIRED, TACK SHALL BE APPLIED AT THE RATE OF 0.08 GALLONS PER SQUARE YARD.

34. UTILITIES BY OTHERS. AT NO ADDITIONAL COST TO THE OWNER, THE CONTRACTOR SHALL PROVIDE ACCESS AND ALLOW FOR COORDINATION SERVICES TO ACCOMMODATE UTILITIES, CONDUIT AND/OR SLEEVE CONSTRUCTION BY OTHERS. ANY UTILITIES, CONDUITS AND/OR SLEEVES INSTALLED BY OTHERS AND DAMAGED BY THE CONTRACTOR SHALL BE REPAIRED AT THE CONTRACTOR'S EXPENSE.

35. CURB TRANSITIONS. ALL EXPOSED ENDS OF CURB SHALL TRANSITION TO FINISHED GRADE

36. CONCRETE. ALL CONCRETE (CURBS, SIDEWALKS, STRUCTURES, ETC.) SHALL BE 3,000 PSI CONCRETE (MINIMUM 28 DAY COMPRESSIVE STRENGTH) UNLESS OTHERWISE SPECIFIED.

37. REINFORCING STEEL. ALL REINFORCING STEEL SHALL CONFORM TO ASTM A-615 AND HAVE A TENSILE STRENGTH OF 60,000 PSI OR AS OTHERWISE NOTED.

38. FIELD REVIEWS. FIELD REVIEWS OF CONSTRUCTED FACILITIES WILL BE REQUIRED DURING THE CONSTRUCTION PROCESS. THE CONTRACTOR IS TO NOTIFY THE ENGINEER A MINIMUM OF 48 HOURS PRIOR TO THE REQUESTED TIME FOR REVIEW OF THE COMPLETED ITEMS. CONTRACTOR TO PROVIDE ALL LABOR, EQUIPMENT AND MATERIALS NECESSARY TO ACCOMMODATE THE FIELD REVIEW. THESE REVIEWS INCLUDE BUT ARE NOT LIMITED TO:

   A. DRAINAGE PIPE AND STRUCTURE REVIEW
   B. WATER/SEWER STRUCTURE, PIPE, AND FITTING REVIEW
   C. SUBGRADE, BASE, & ASPHALT REVIEW
   D. CURB & SIDEWALK REVIEWS
   E. PUMP OUT AND LAMPING OF DRAINAGE SYSTEM
   F. SEMI-FINAL AND FINAL REVIEWS

42. COMPLETENESS FOR REVIEW. FACILITIES MUST BE COMPLETED PRIOR TO SCHEDULING THE FIELD REVIEW. IF A FIELD REVIEW FAILS TO MEET THE AGENCY REQUIREMENTS AND AN ADDITIONAL REVIEW IS REQUIRED, THE CONTRACTOR WILL BE RESPONSIBLE TO PAY ANY FEES ASSOCIATED WITH THE ADDITIONAL REVIEW.

43. TESTING. TESTING OF CONSTRUCTION MATERIALS AND/OR CONSTRUCTED FACILITIES WILL BE REQUIRED AS OUTLINED IN THE GOVERNING SPECIFICATIONS AS NECESSARY TO MEET THE REQUIREMENTS OF THE JURISDICTIONAL AGENCIES. ALL TESTING IS TO BE DONE BY A FLORIDA LICENSED TESTING LABORATORY. THE CONTRACTOR IS TO SUBMIT ALL REQUIRED TESTS ON UTILITY EMBANKMENT, BACKFILL, SUBGRADE, BASE AND SURFACE COURSE MATERIAL TO THE ENGINEER PRIOR TO ANY REQUEST FOR PAYMENT. IF ANY TESTS FAIL TO MEET THE SPECIFICATIONS, THE CONTRACTOR, AT ITS EXPENSE, SHALL CORRECT ALL DEFICIENT WORK AND SUBMIT TEST RESULTS INDICATING COMPLIANCE WITH THESE SPECIFICATIONS PRIOR TO ANY REQUEST FOR PAYMENT.

44. DENSITY TESTING. THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING THE NECESSARY TESTING TO ASSURE THAT THE PROPER COMPACTION HAS BEEN ACHIEVED ON THE SUBGRADE, BASE MATERIAL, PIPE BASE MATERIAL AND ALL OTHER PERTINENT AREAS THAT HAVE BEEN COMPLETED. THE CONTRACTOR SHALL BEAR ALL COSTS ASSOCIATED WITH TESTING AND RETESTING THE AREAS AND SHALL PROVIDE THE OWNER AND THE ENGINEER COPIES OF ALL TEST RESULTS. ALL REPORTS ARE TO BE SIGNED AND SEALED BY A REGISTERED GEOTECHNICAL ENGINEER IN THE STATE OF FLORIDA.

45. HORIZONTAL DISTRIBUTION OF DENSITY TESTING. DENSITY TESTING SHALL BE PERFORMED AT THE FOLLOWING MINIMUM LOCATIONS WITH HORIZONTAL SPACING PER THE VERTICAL DISTRIBUTION OF DENSITY TAKING NOTE. MORE SPECIFICALLY:

   A. PIPE TRENCHES – TESTS SHALL BE PERFORMED AT RANDOMLY SELECTED LOCATIONS WITHIN EACH 300 FOOT INTERVAL (MAXIMUM)ALONG THE LENGTH OF A PIPE INSTALLATION, AND BETWEEN EACH SET OF STRUCTURES SEPARATED BY LESS THAN 300 FEET. AT LEAST ONE TEST SHALL BE PERFORMED FOR EVERY 12 INCH BACKFILL LAYER, STARTING WITH THE FIRST TEST PERFORMED AT THE PIPE SPRINGLINE (COVERING THE 12" LAYER BELOW SPRINGLINE ELEVATION).

   B. STRUCTURES – TESTS SHALL BE PERFORMED AT EACH MANHOLE OR DRAINAGE STRUCTURE. AT LEAST ONE TEST SHALL BE PERFORMED FOR EVERY 12 INCH BACKFILL LIFT, STARTING WITH THE FIRST TEST PERFORMED AT THE BASE OF THE STRUCTURE (COVERING THE 12" LAYER BELOW BASE ELEVATION).

   C. SUBGRADE – TESTS SHALL BE PERFORMED AT RANDOMLY SELECTED LOCATIONS WITHIN EACH 300 FOOT INTERVAL (MAXIMUM) ALONG THE LENGTH OF THE ROADWAY, SIDEWALK, OR PATHWAY AND EVERY 6,000 SQUARE FEET OF PARKING AREA.

   D. BASE ROCK – TESTS SHALL BE PERFORMED AT RANDOMLY SELECTED LOCATIONS WITHIN EACH 300 FOOT INTERVAL (MAXIMUM) ALONG THE LENGTH OF THE ROADWAY, SIDEWALK, OR PATHWAY AND EVERY 6,000 SQUARE FEET OF PARKING AREA.

   E. ROAD CROSSINGS – TESTS SHALL BE PERFORMED FOR EACH 12 INCH LIFT WITHIN ANY OPEN CUT ROAD CROSSING LOCATION.

47. TACTILE SURFACES – ALL TACTILE SURFACES SHALL COMPLY WITH ALL APPLICABLE CODES AND REQUIREMENTS INCLUDING BUT NOT LIMITED TO, THE FLORIDA DEPARTMENT OF TRANSPORTATION (FDOT) INDEX 304 AND THE FLORIDA BUILDING CODE (FBC), ACCESSIBILITY.

ABBREVIATIONS

BURIED FIBER OPTIC	BFO
PROPERTY LINE	P/L
EXISTING	EXIST. / EX
FIRE HYDRANT	FH
GATE VALVE	GV
LINEAR FOOT	LF
MANHOLE	MH
RIGHT-OF-WAY	R/W
SANITARY	SAN
STORM	STM
STORM MANHOLE	SMH
UTILITY EASEMENT	U.E.
UNKNOWN	UNK
WATER	WTR
WATERMAIN	WM

LEGEND

EXISTING WATER MAIN	EXWM
EXISTING SANITARY SEWER	EXSS
EXISTING FORCE MAIN	EXFM
EXISTING STORM DRAINAGE	EXDN
EXISTING GAS LINE	GAS
EXISTING OVERHEAD WIRES	OH
EXISTING EDGE OF PAVEMENT	

ALL ELEVATIONS REFER TO NORTH AMERICAN VERTICAL DATUM OF 1988 (NAVD 88).
TO CONVERT TO NGVD 29 DATUM, ADD 1.54' TO NAVD ELEVATIONS.

NOTE: THESE PLANS ARE PRELIMINARY AND SHOULD NOT BE USED FOR CONSTRUCTION OR BIDDING/CONTRACTUAL PURPOSES UNTIL ALL PERMITS ARE ISSUED.

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DRAWN: P. Saffold
DESIGNED: TNM
APPROVED: TNM
PROJECT #21-030

NO.	DATE	REVISIONS

GENERAL NOTES & SPECIFICATIONS

1319 MANGONIA PARK WAREHOUSE
[redacted]
MANGONIA PARK, FLORIDA

TOWN RESUBMITTAL
DATE: 8/1/2024

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